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Jeon et al.

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(54) **METHOD FOR ALIGNING GRADIENT SYMBOLS BY USING BIAS REGARDING AIRCOMP IN SIGNAL AMPLITUDE RANGE OF RECEIVER**

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See application file for complete search history.

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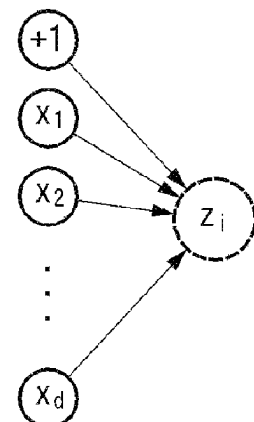
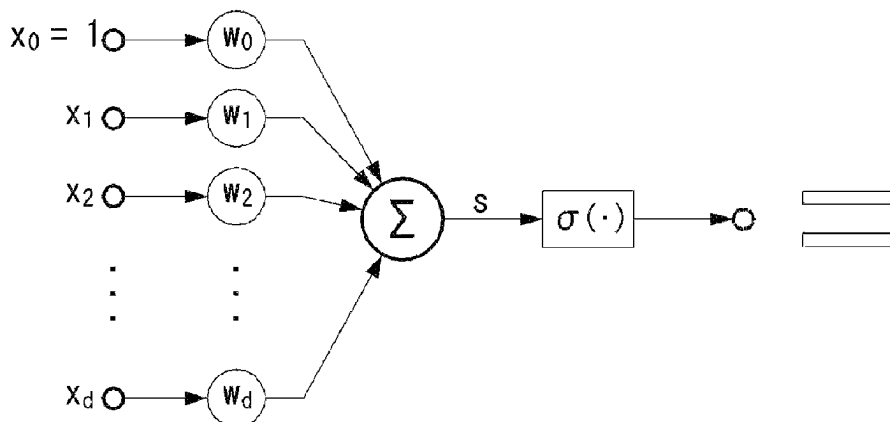
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(57) **ABSTRACT**

Disclosed is a method for aligning gradient symbols by using bias regarding AirComp in a signal amplitude range of a receiver. A method for aligning gradient symbols according to an embodiment of the present disclosure comprises the steps of: performing iteration of gradient values included in a first planar vector and applying clipping and bias values so as to obtain a second planar vector having gradient values in which symbols are aligned; and transmitting the second planar vector together with channel information to a server in the form of AirComp. Various embodiments of the present disclosure may be linked to an artificial intelligence module, a drone (unmanned aerial vehicle, UAV), a robot, an augmented reality (AR) device, a virtual reality (VR) device, a device related to a 5G service, and the like.

12 Claims, 17 Drawing Sheets



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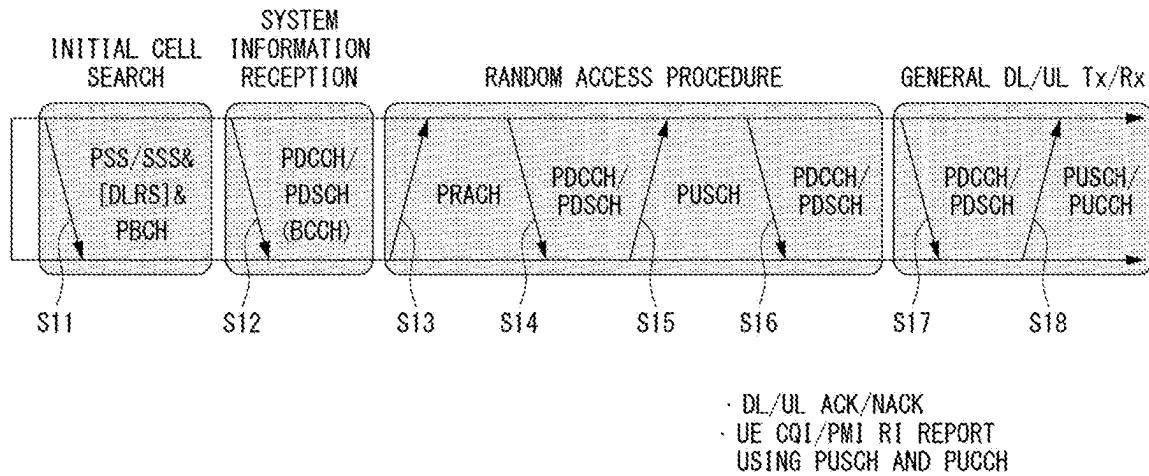
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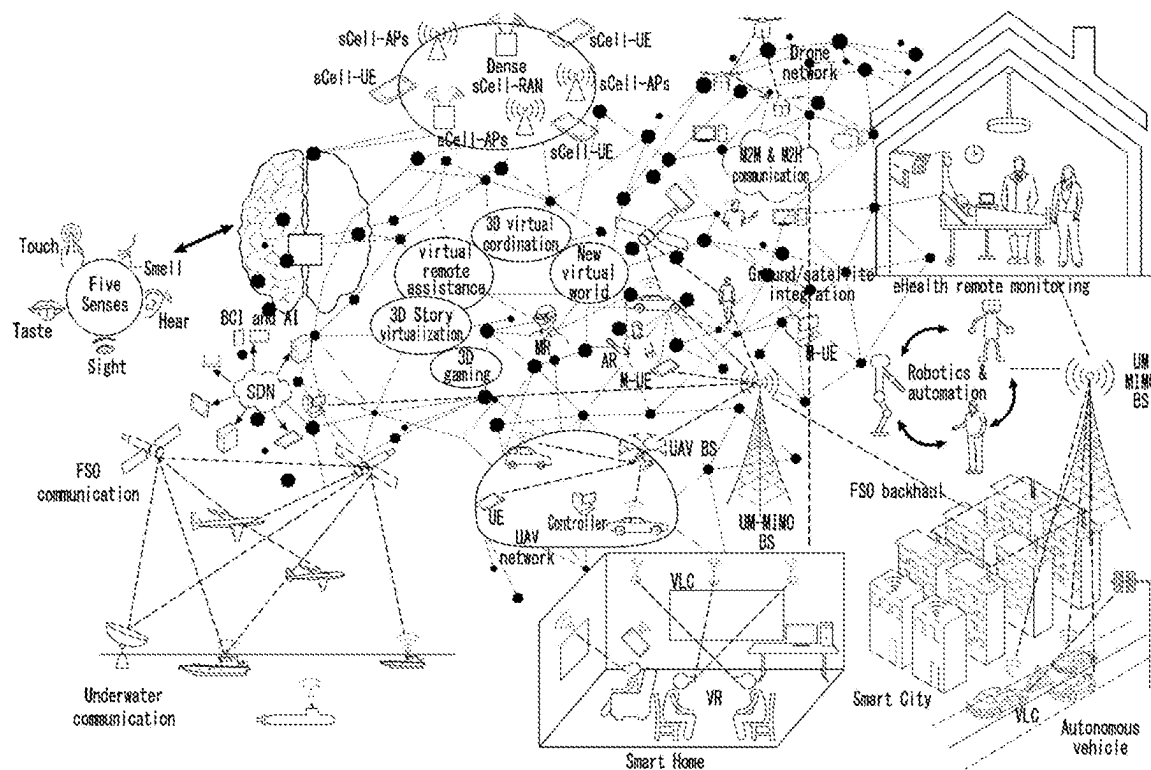
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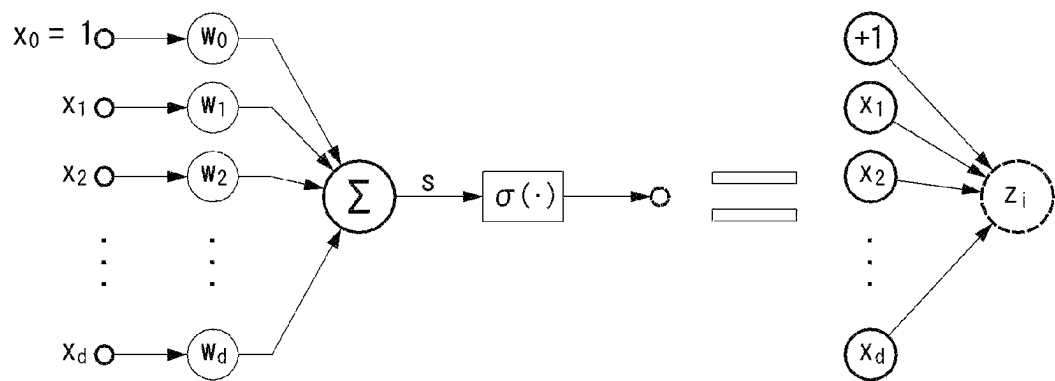
[FIG. 1]



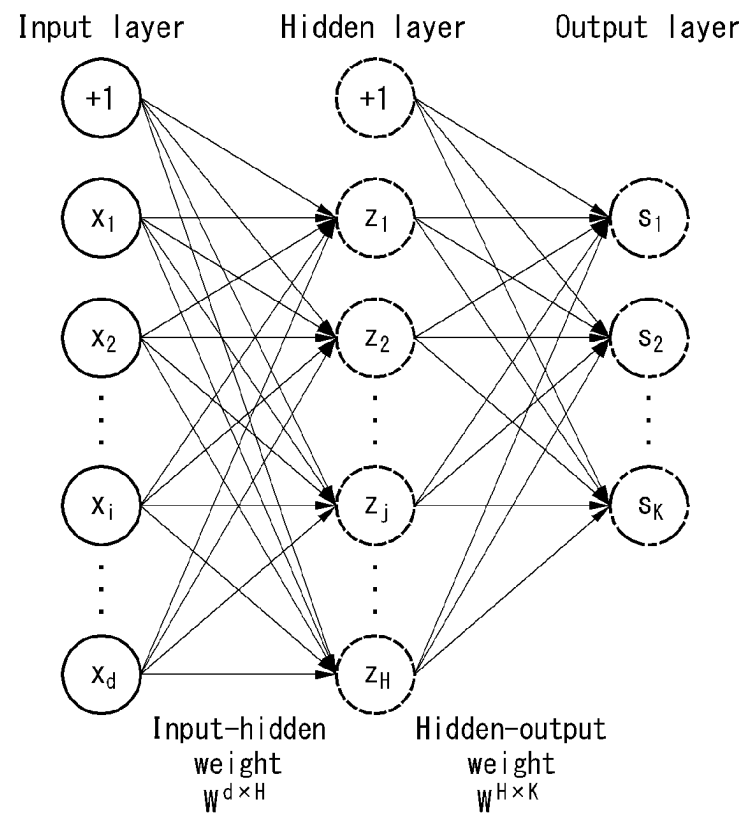
[FIG. 2]



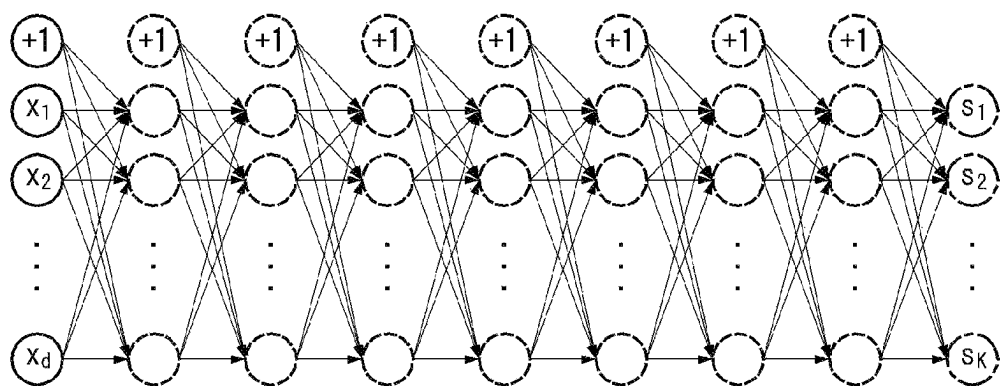
[FIG. 3]



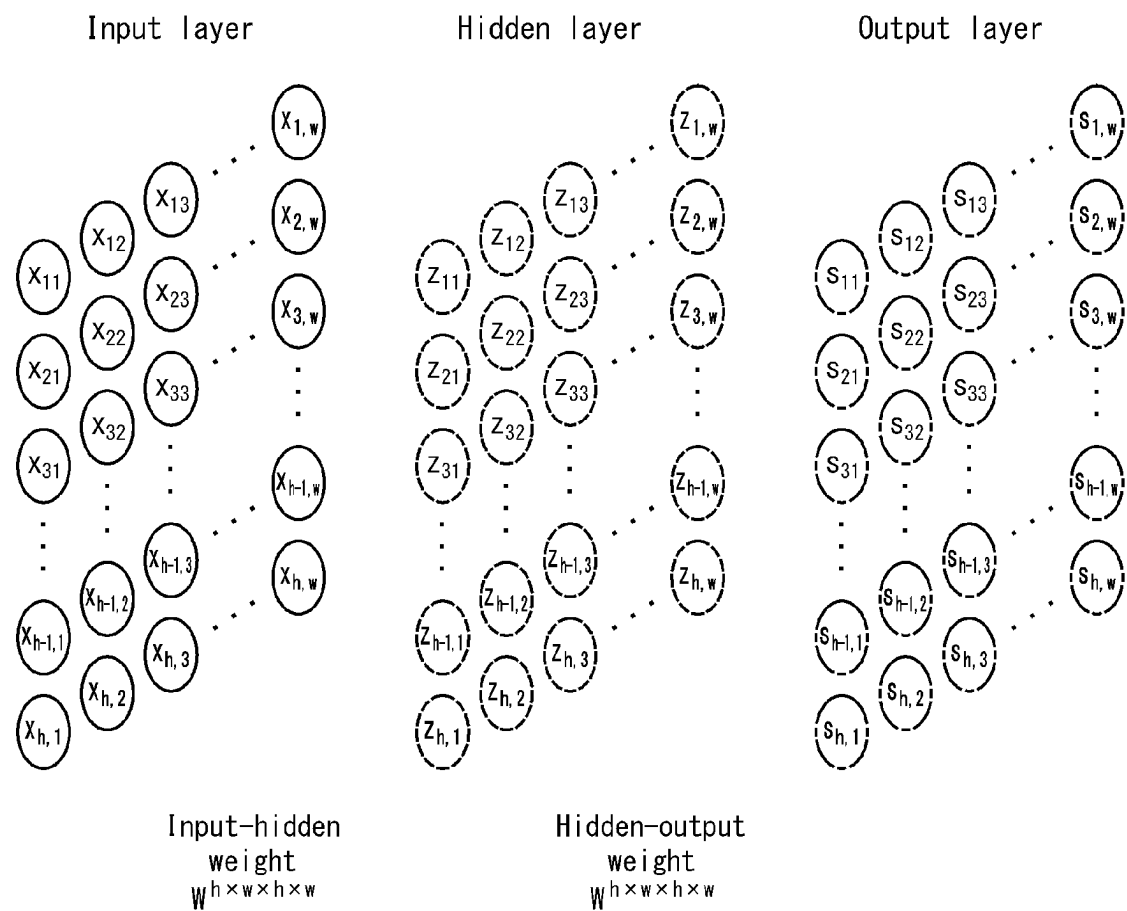
[FIG. 4]



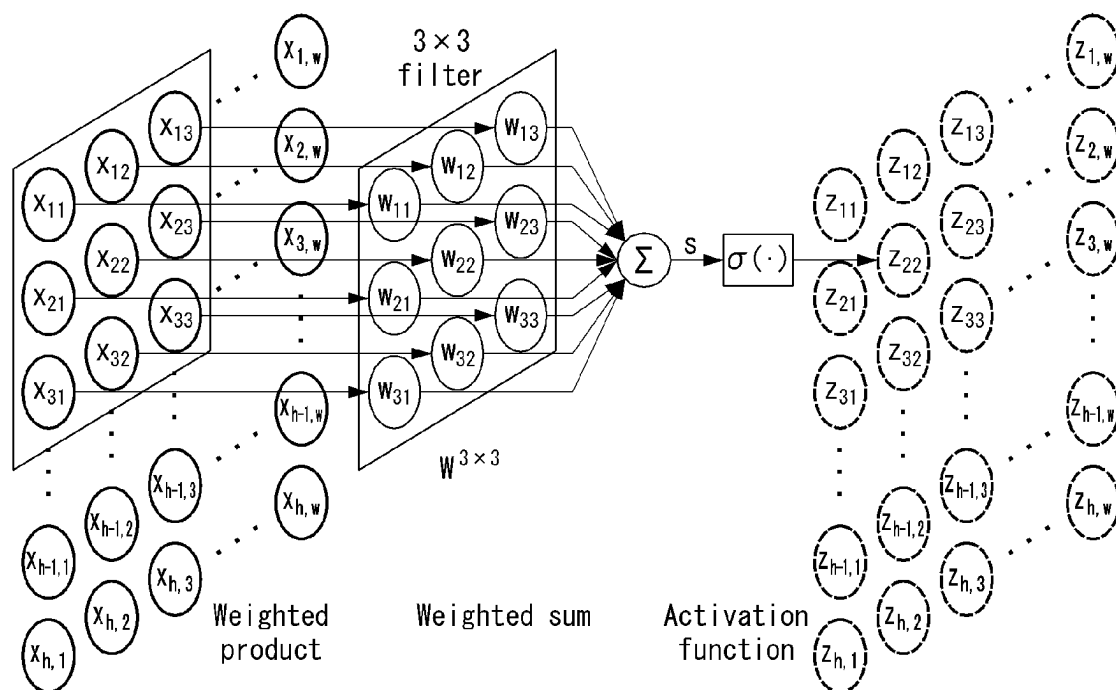
[FIG. 5]



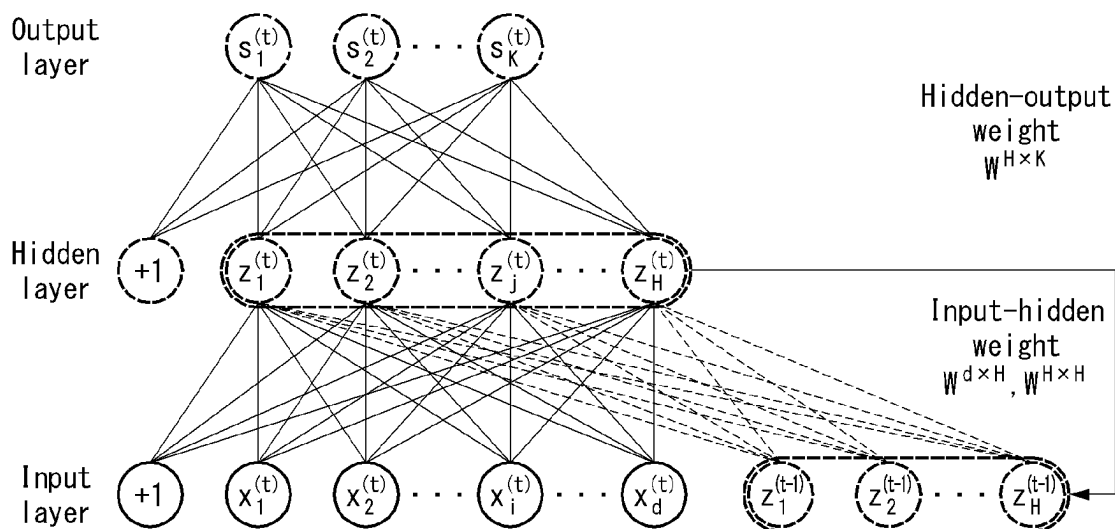
[FIG. 6]



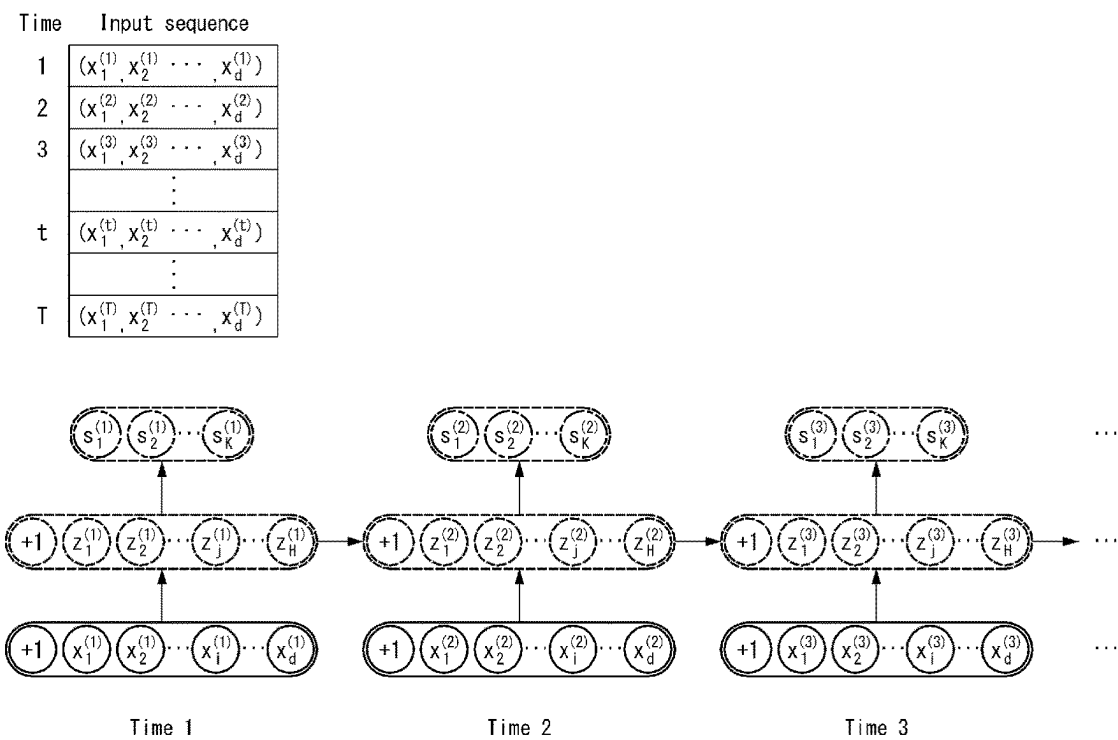
[FIG. 7]



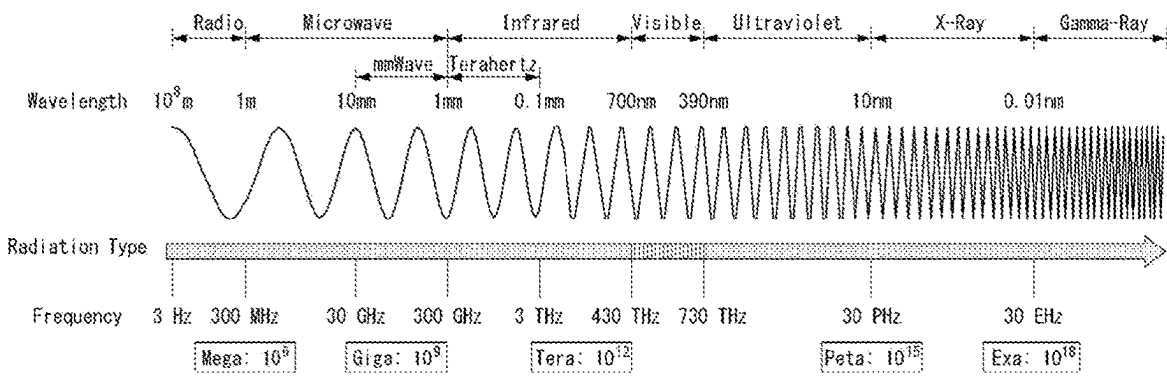
[FIG. 8]



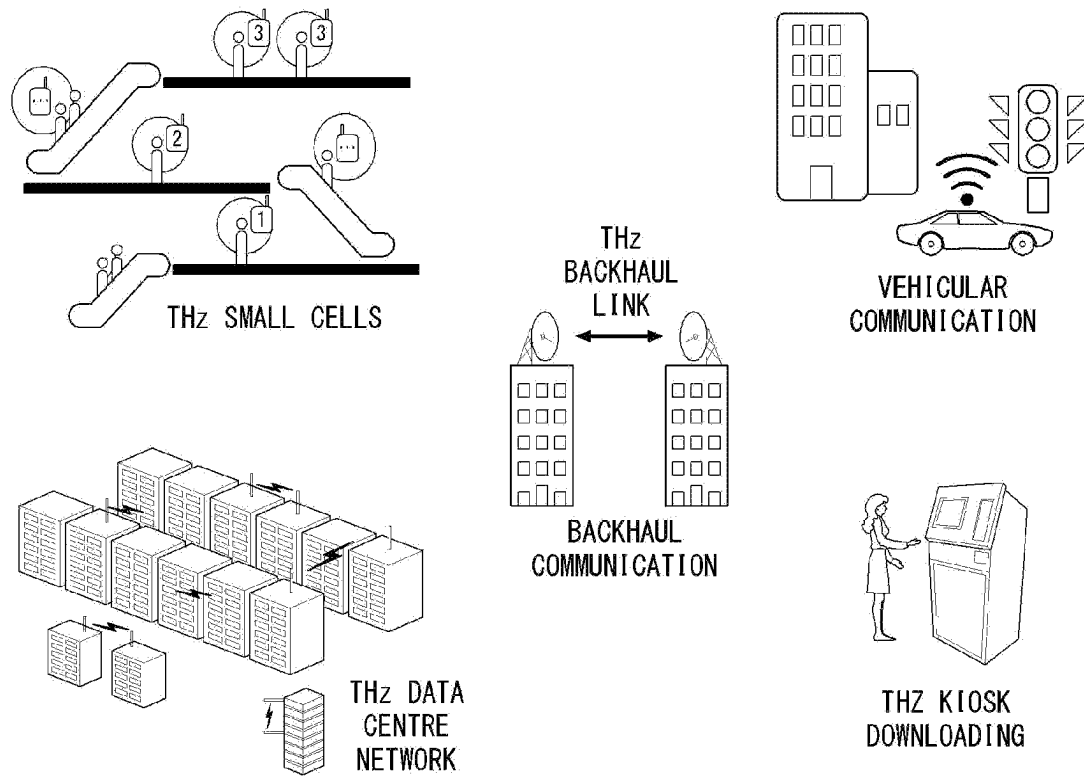
[FIG. 9]



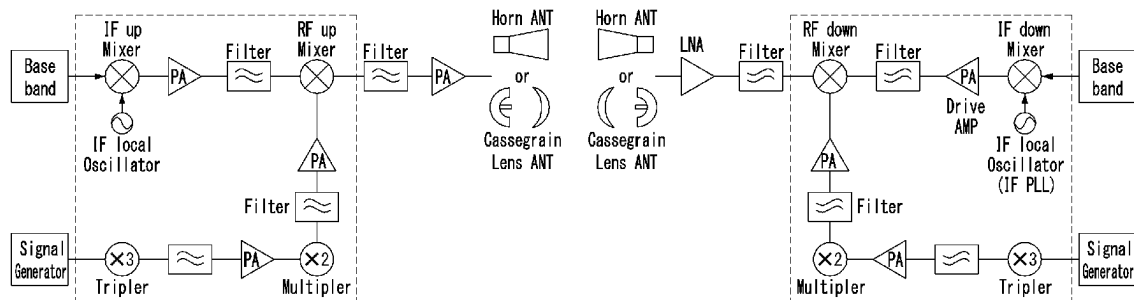
[FIG. 10]



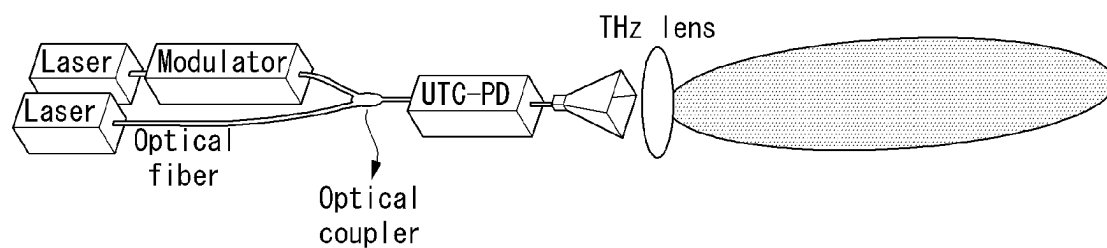
[FIG. 11]



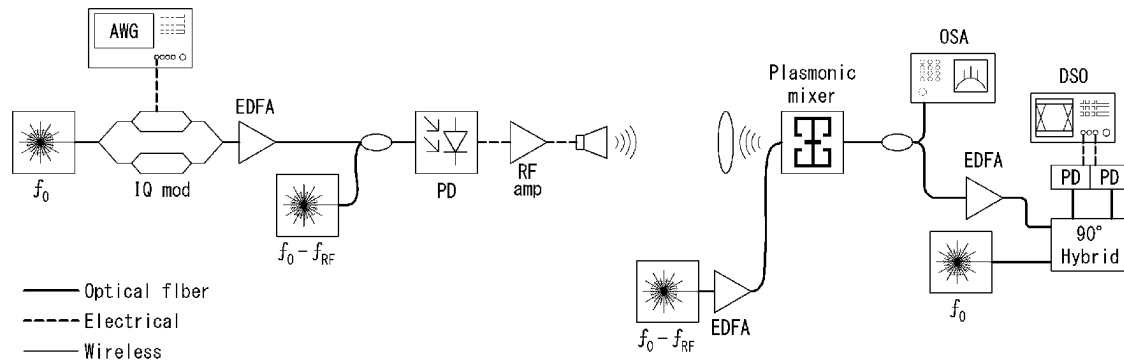
[FIG. 12]



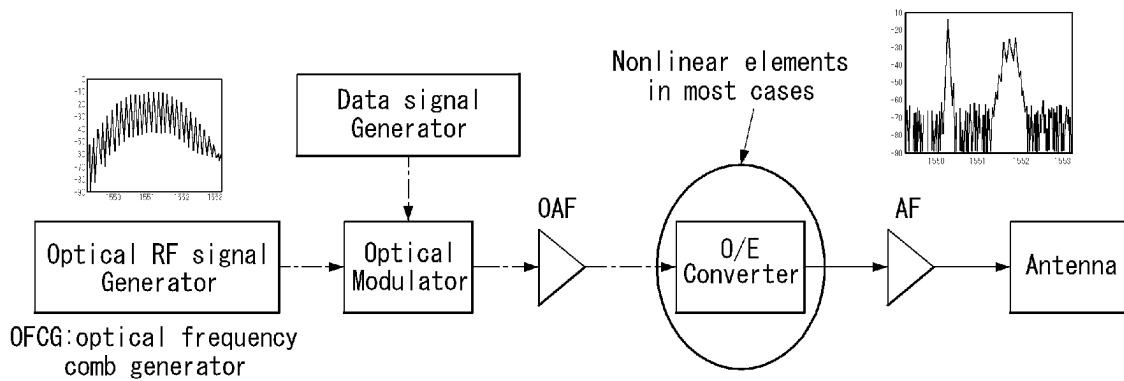
[FIG. 13]



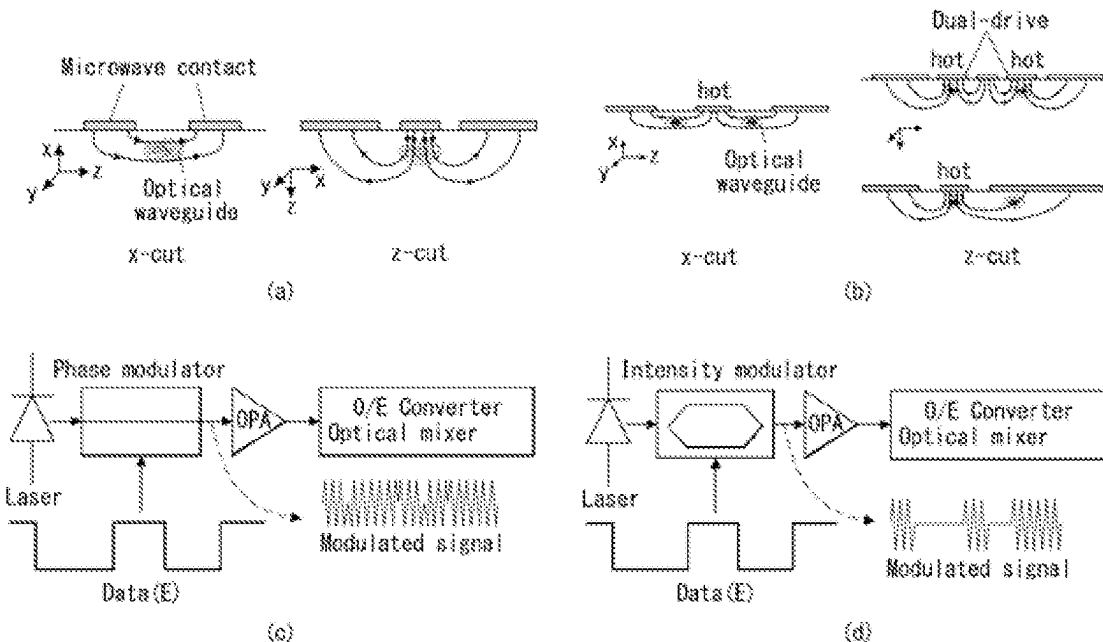
[FIG. 14]



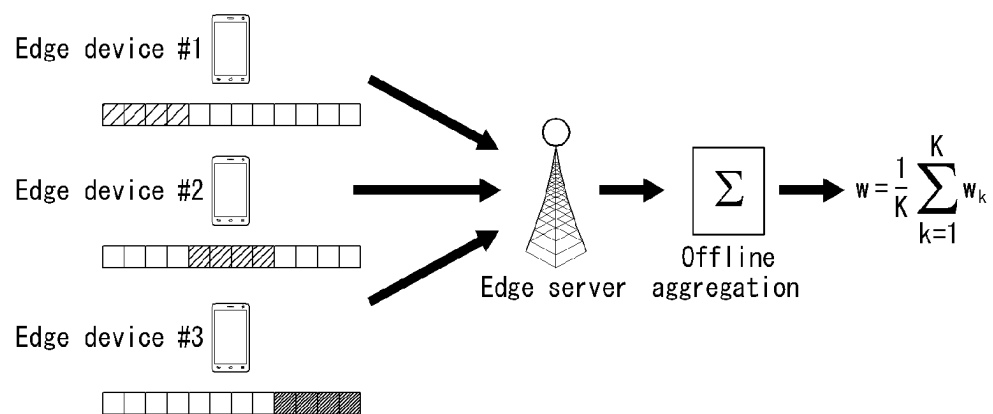
[FIG. 15]



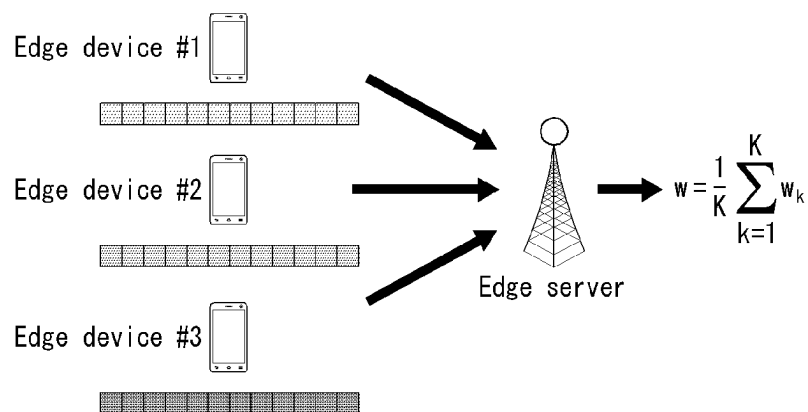
[FIG. 16]



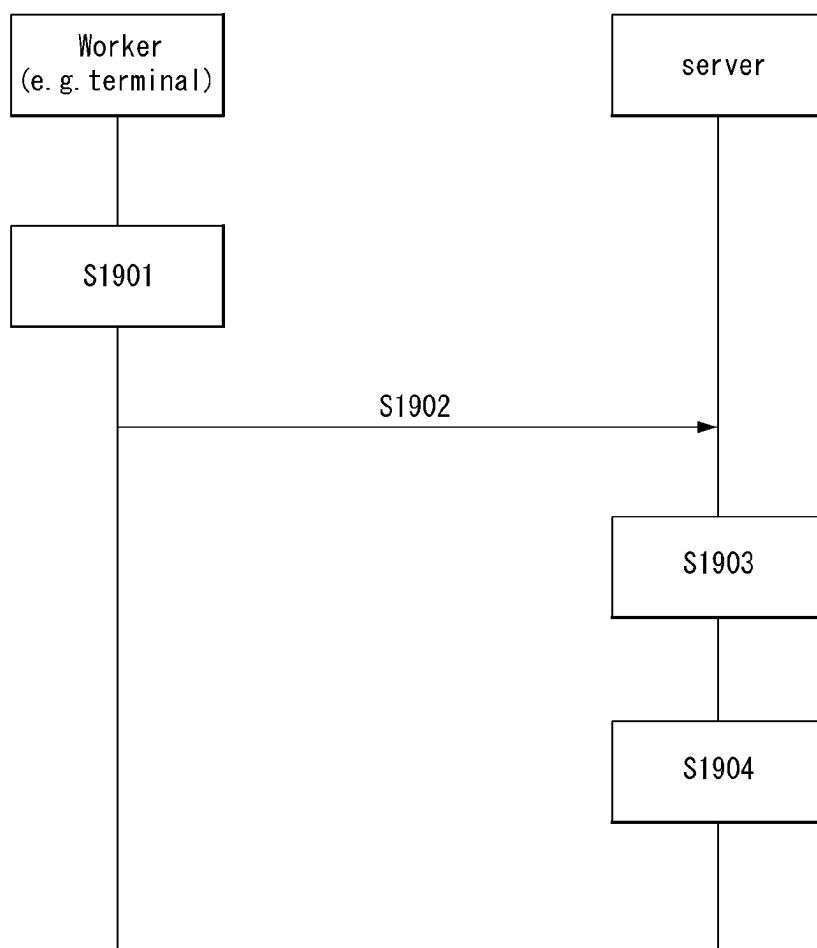
[FIG. 17]



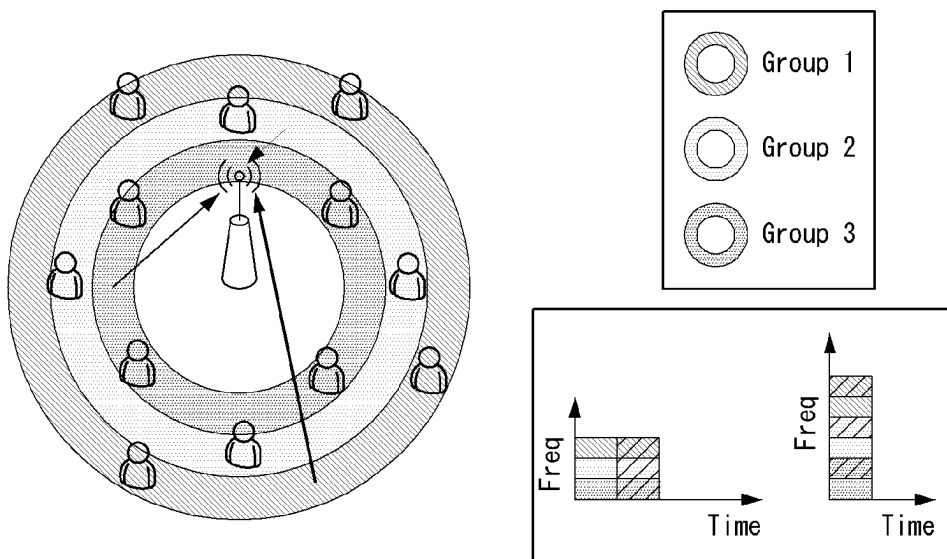
[FIG. 18]



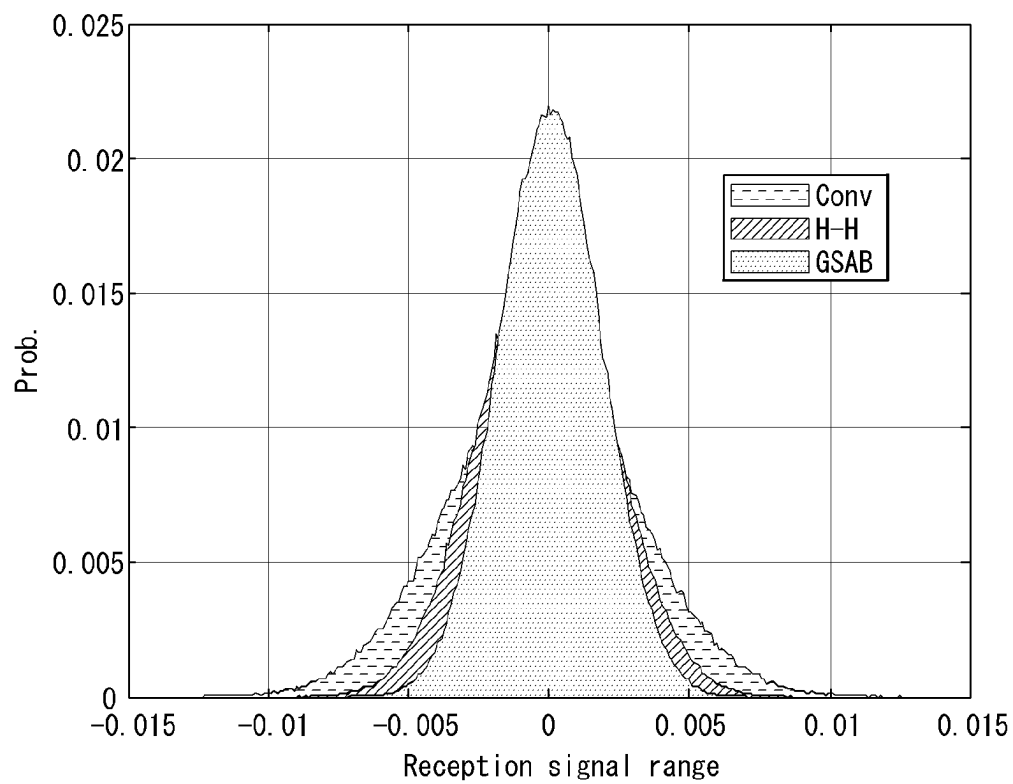
[FIG. 19]



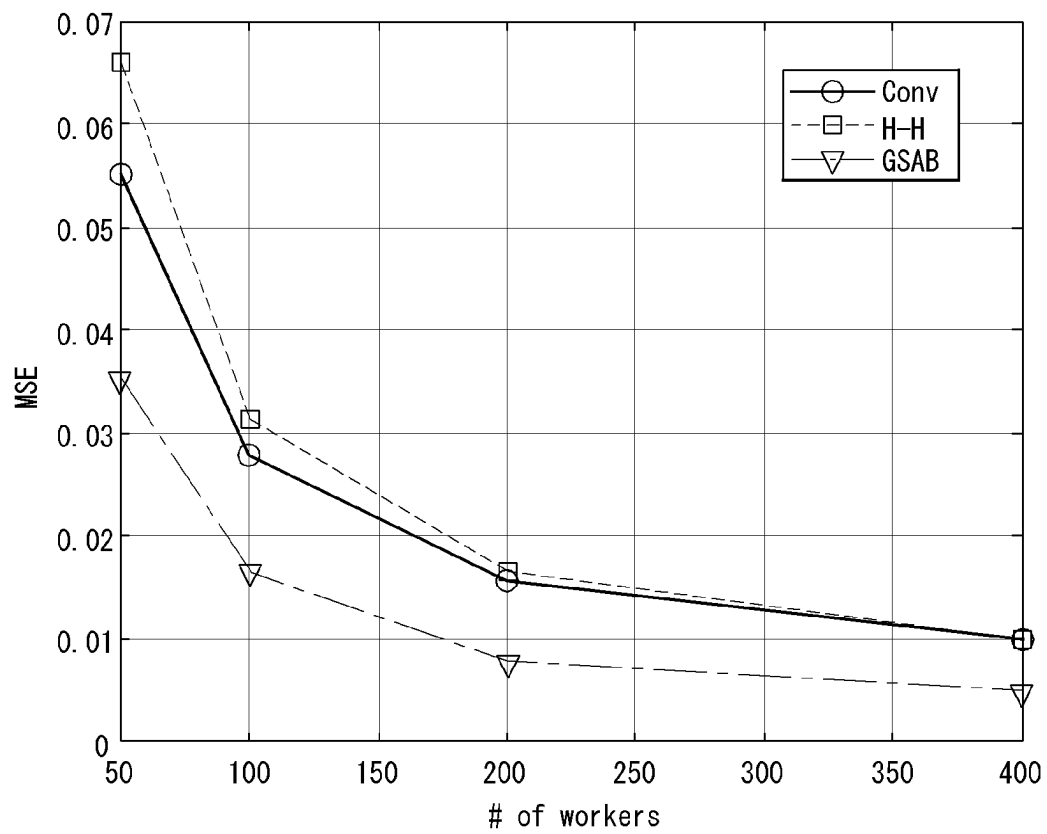
[FIG. 20]



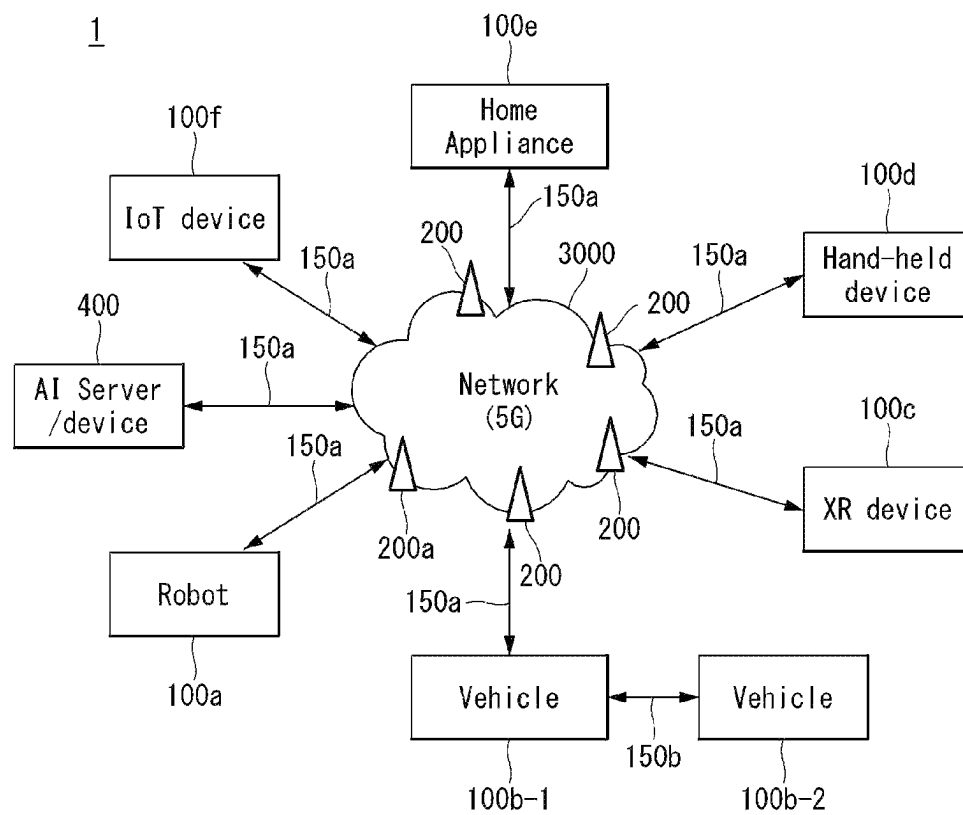
[FIG. 21]



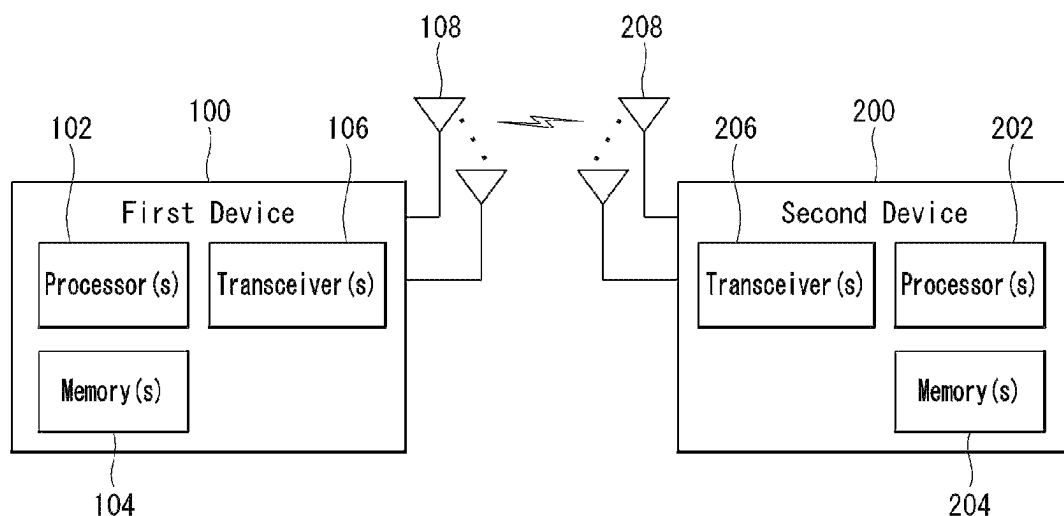
[FIG. 22]



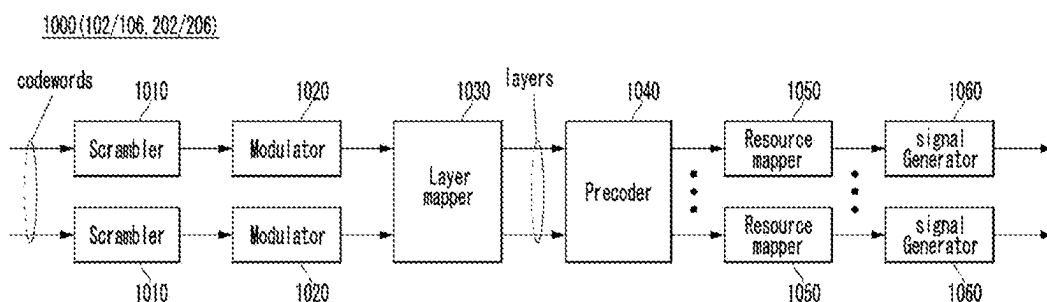
[FIG. 23]



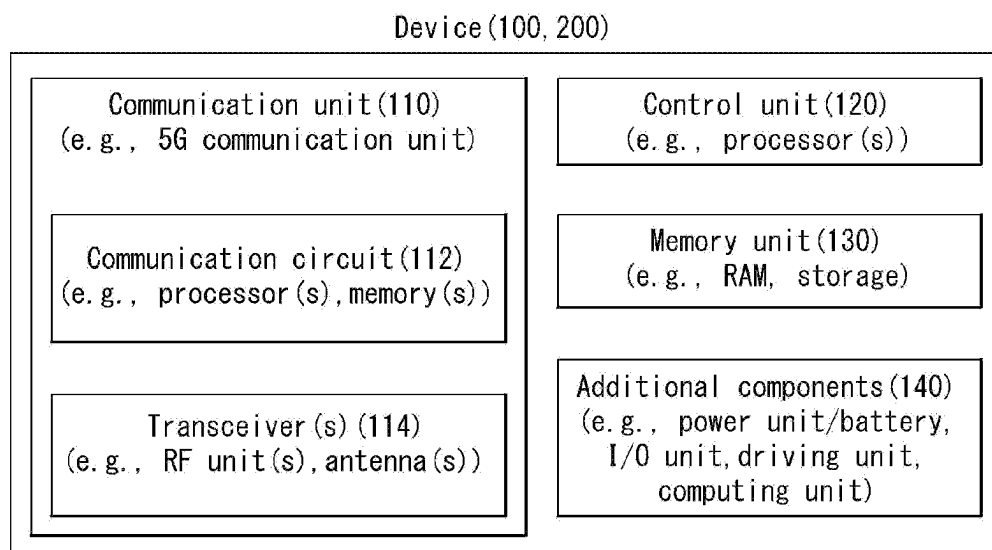
[FIG. 24]



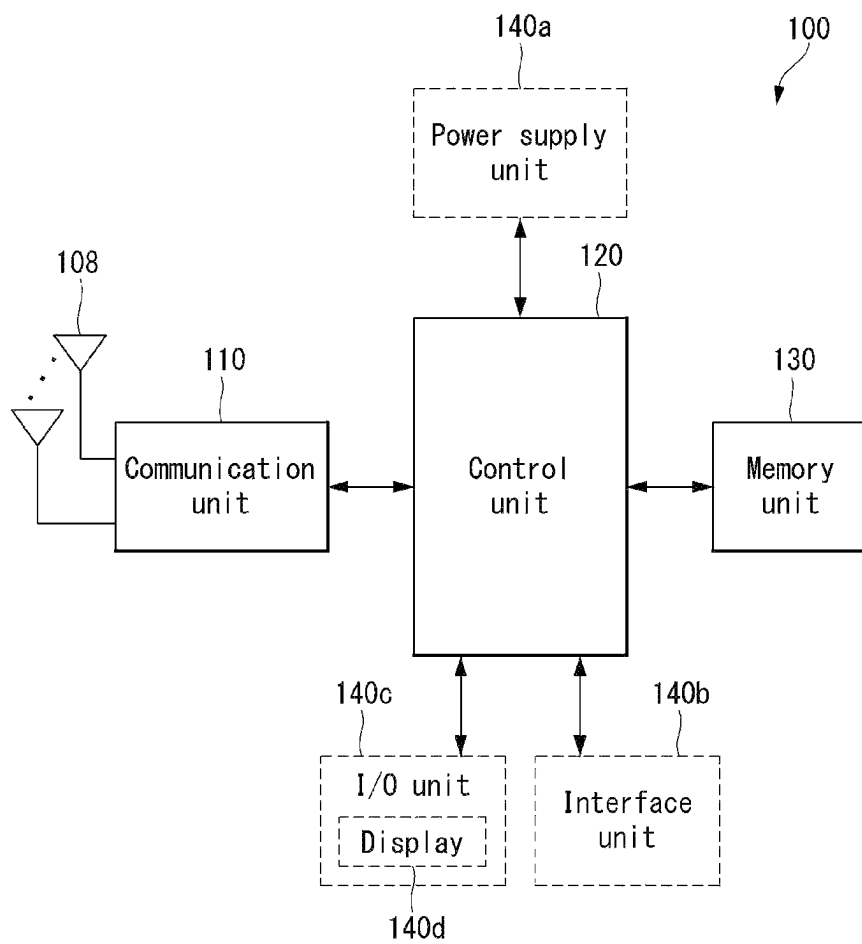
[FIG. 25]



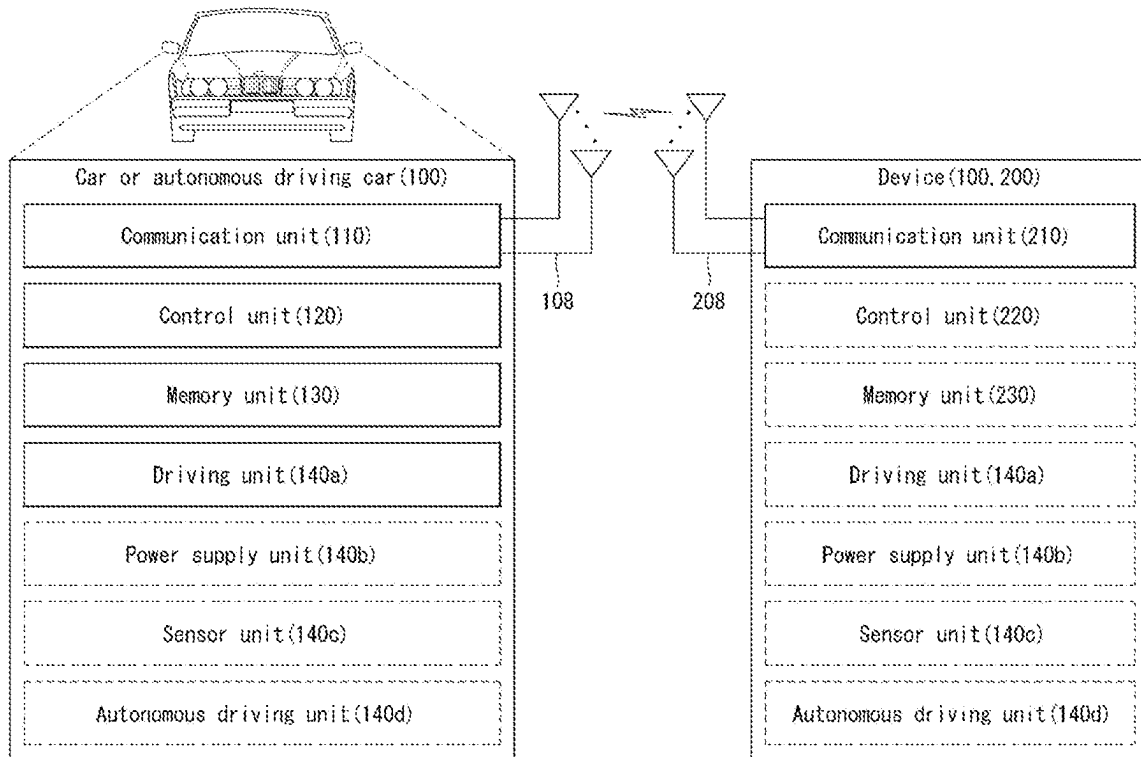
[FIG. 26]



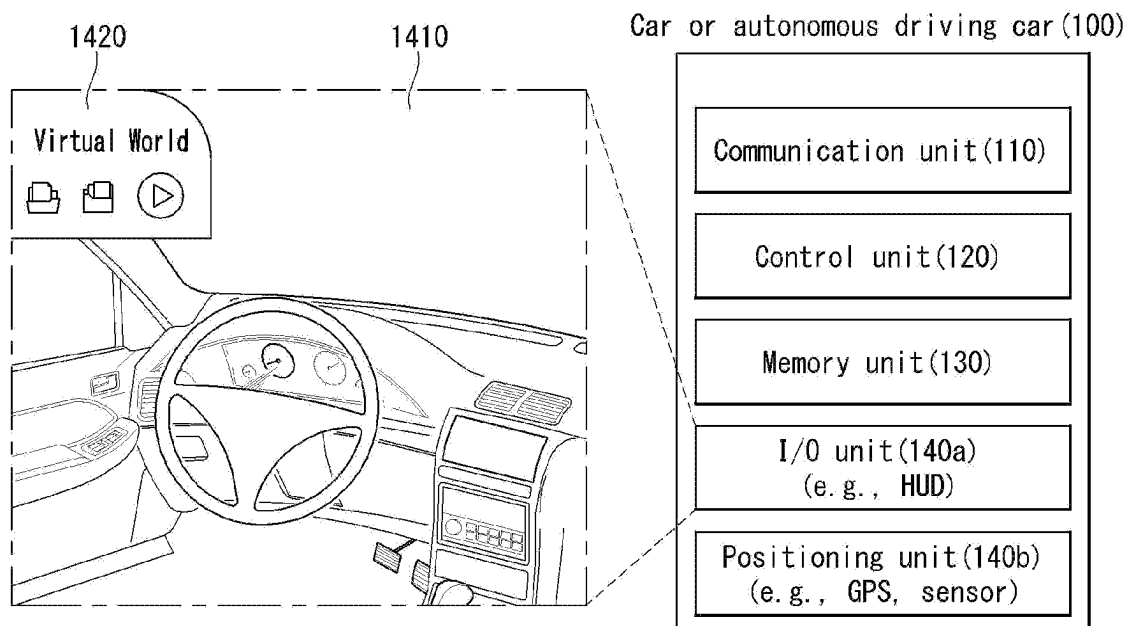
[FIG. 27]



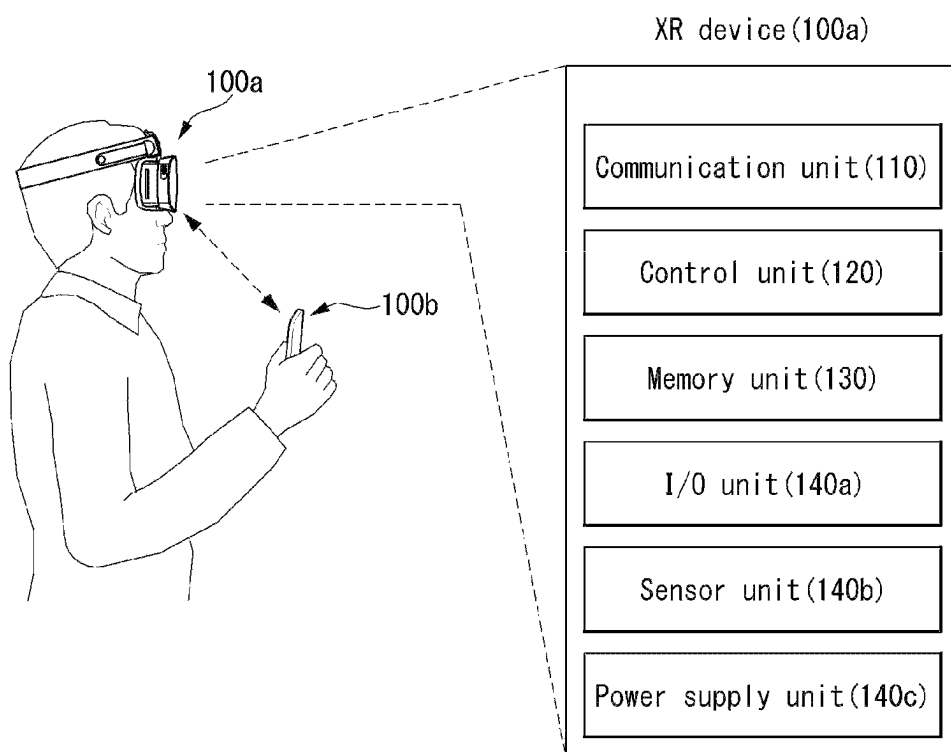
[FIG. 28]



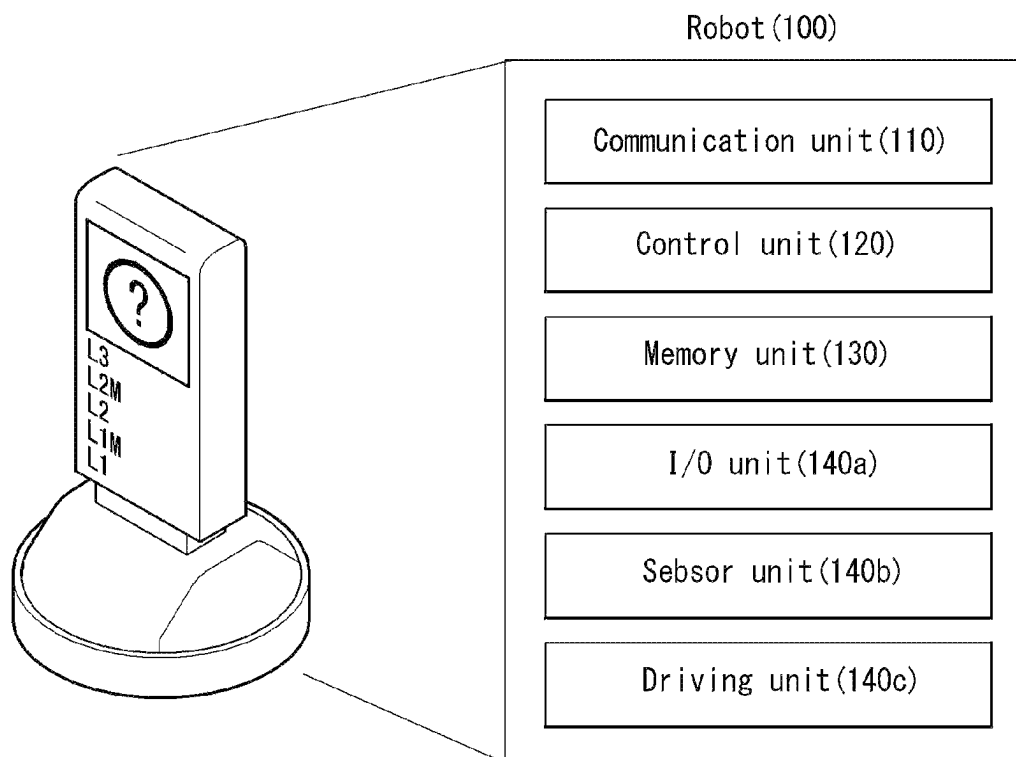
[FIG. 29]



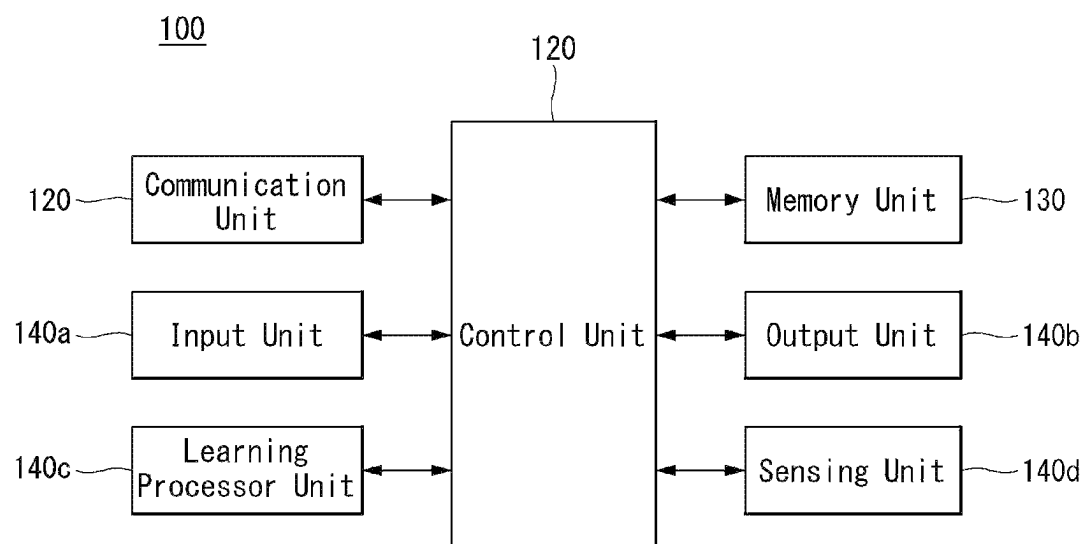
[FIG. 30]



[FIG. 31]



[FIG. 32]



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METHOD FOR ALIGNING GRADIENT SYMBOLS BY USING BIAS REGARDING AIRCOMP IN SIGNAL AMPLITUDE RANGE OF RECEIVER

CROSS-REFERENCE TO RELATED APPLICATION

This application is the National Stage filing under 35 U.S.C. 371 of International Application No. PCT/KR2020/012060, filed on Sep. 7, 2020, the contents of which are all hereby incorporated by reference herein in their entirety.

TECHNICAL FIELD

The present disclosure relates to a method of aligning gradient signs using a bias for AirComp in a range of a signal amplitude of a receiver, and more specifically, a method of aligning gradient signs using a bias for AirComp over a possible range of a signal amplitude in a receiver.

BACKGROUND ART

Wireless communication systems are widely deployed to provide various types of communication services such as voice and data, and attempts to incorporate AI into a communication system are rapidly increasing. The attempted AI incorporation schemes can be greatly divided into communications for AI (C4AI) for developing a communication technology to support artificial intelligence (AI), and AI for communications (AI4C) utilizing AI for improvement of communication performance. In an AI4C area, there is an attempt to increase design efficiency by replacing a channel encoder/decoder with an end-to-end autoencoder. In a C4AI area, there is a method of updating a common prediction model while protecting personal information by sharing only a weight or gradient of a model with a server without sharing device raw data through federated Learning, which is one distributed learning scheme. There is a method of distributing loads of a device, a network edge, and a cloud server through split inference.

DISCLOSURE

Technical Problem

An object of the present disclosure is to solve the aforementioned needs and/or problems.

Another object of the present disclosure is to implement a method of aligning gradient signs using a bias for AirComp in a range of a signal amplitude of a receiver in which enabling transmission and reception without a loss of power or a loss of the number of devices participating in learning at the time of federated learning can intuitively narrow a received signal distribution.

Technical Solution

According to an embodiment of the present disclosure, a method, by a terminal, of aligning gradient signs in a wireless communication system for federated learning, wherein the terminal includes a local model and a server includes a global model, the method comprising: obtaining a second flatten vector having gradient values in which signs are aligned, by repeating gradient values included in a first flatten vector and applying clipping and bias values; and

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transmitting the second flatten vector together with channel information to the server in the form of AirComp.

Furthermore, wherein the bias value is determined by the number of terminals participating in federated learning and statistical characteristics of the gradients of the first flatten vector.

Furthermore, wherein the bias value is determined using a value defined in advance as a table in a transceiver.

Furthermore, wherein the channel information is subjected to phase compensation.

Furthermore, wherein the signs include a positive sign and a negative sign, values having a negative sign in a first partial vector among the gradient values included in the second flatten vector are clipped, and a negative bias is applied to the partial vector, and values having a positive sign in a second partial vector among the gradient values included in the second flatten vector are clipped, and a positive bias is applied to the partial vector.

Furthermore, wherein a signal transmitted in the transmitting in the form of AirComp has a predetermined transmission power.

Furthermore, further comprising: grouping at least one terminal participating in learning among a plurality of random terminals based on CSI information.

Furthermore, wherein a signal transmitted by at least one terminal belonging to the same group has contiguous reception power sensitivity.

According to another embodiment of the present disclosure, a method of aligning gradient signs using a server in a wireless communication system for federated learning wherein a terminal includes a local model and the server includes a global model, the method comprising: receiving gradient signals for federated learning transmitted in an AirComp form from a plurality of terminals, the signal includes channel information and a second flatten vector; obtaining an aggregated gradient based on the received gradient signals; and updating parameters of the global model based on the aggregated gradient.

Furthermore, wherein the obtaining of the aggregated gradient includes calculating one aggregated gradient by overlapping the gradient signals for each entry.

Furthermore, wherein a bias value is eliminated and not present in the aggregated gradient as a result of overlapping for each entry.

Furthermore, a computer system-readable recording medium having a program for executing the method in a computer system recorded thereon.

Advantageous Effects

Effects of the method of aligning gradient signs using a bias for AirComp in a range of a signal amplitude of a receiver according to an embodiment of the present disclosure will be described.

In the present disclosure, enabling transmission and reception without a loss of power or a loss of the number of devices participating in learning at the time of federated learning can intuitively narrow a received signal distribution.

Effects that can be obtained in the present disclosure are not limited to the effects mentioned above, and other effects not mentioned will be clearly understood by those skilled in the art from the following description.

DESCRIPTION OF DRAWINGS

FIG. 1 illustrates physical channels and general signal transmission used in a 3GPP system.

FIG. 2 is a diagram illustrating an example of a communication structure providable in a 6G system.

FIG. 3 illustrates a perceptron structure.

FIG. 4 illustrates a multi-perceptron structure.

FIG. 5 illustrates a deep neural network structure.

FIG. 6 illustrates a convolutional neural network structure.

FIG. 7 illustrates a filter operation in a convolutional neural network.

FIG. 8 illustrates a neural network structure in which a recurrent loop is present.

FIG. 9 illustrates an operating structure of a recurrent neural network.

FIG. 10 illustrates an example of an electromagnetic spectrum.

FIG. 11 illustrates an example of THz communication application.

FIG. 12 illustrates an example of an electronic element-based THz wireless communication transceiver.

FIG. 13 illustrates an example of a method of generating a THz signal based on an optical element.

FIG. 14 illustrates an example of an optical element-based THz wireless communication transceiver.

FIG. 15 illustrates a photonic source-based transmitter structure.

FIG. 16 illustrates a structure of an optical modulator.

FIG. 17 is a diagram schematically illustrating OFDMA-based federated learning.

FIG. 18 is a diagram schematically illustrating AirComp-based federated learning.

FIG. 19 is a sequence diagram of a method of aligning gradient signs according to an embodiment of the present disclosure.

FIG. 20 is a diagram simply illustrating a resource management method for a proposed scheme.

FIG. 21 is a diagram illustrating a received signal distribution when transmission is performed with the same transmission power P in all of a traditional scheme, an H-H scheme, and a GSAB scheme.

FIG. 22 is a diagram illustrating a performance comparison between respective schemes when it is assumed that a range of aggregate received signals is $[-1, 1]$.

FIG. 23 illustrates a communication system applied to the present disclosure.

FIG. 24 illustrates a wireless device that can be applied to the present disclosure.

FIG. 25 illustrates a signal processing circuit for a transmission signal.

FIG. 26 shows another example of a wireless device applied to the present disclosure.

FIG. 27 illustrates a portable device applied to the present disclosure.

FIG. 28 illustrates a vehicle or an autonomous vehicle to which the present disclosure is applied.

FIG. 29 illustrates a vehicle applied to the present disclosure.

FIG. 30 illustrates an XR device applied to the present disclosure.

FIG. 31 illustrates a robot applied to the present disclosure.

FIG. 32 illustrates an AI device applied to the present disclosure.

The accompanying drawings, which are included to provide a further understanding of the present disclosure and constitute a part of the detailed description, illustrate

embodiments of the present disclosure and together with the description serve to explain the principle of the present disclosure.

MODE FOR DISCLOSURE

Hereinafter, an embodiment disclosed in the present disclosure will be described in detail with reference to the accompanying drawings and the same or similar components are denoted by the same reference numerals regardless of a sign of the drawing, and duplicated description thereof will be omitted. Suffixes “module” and “unit” for components used in the following description are given or mixed in consideration of easy preparation of the present disclosure only and do not have their own distinguished meanings or roles. Further, in describing an embodiment disclosed in the present disclosure, a detailed description of related known technologies will be omitted if it is determined that the detailed description makes the gist of the embodiment of the present disclosure unclear. Further, it is to be understood that the accompanying drawings are just used for easily understanding the embodiments disclosed in the present disclosure and a technical spirit disclosed in the present disclosure is not limited by the accompanying drawings and all changes, equivalents, or substitutes included in the spirit and the technical scope of the present disclosure are included.

Terms including an ordinary number, such as first and second, are used for describing various elements, but the elements are not limited by the terms. The terms are used only to discriminate one element from another element.

It should be understood that, when it is described that a component is “connected to” or “accesses” another component, the component may be directly connected to or access the other component or a third component may be present therebetween. In contrast, when it is described that a component is “directly connected to” or “directly accesses” another component, it is understood that no element is present between the element and another element.

A singular form includes a plural form if there is no clearly opposite meaning in the context.

In the present application, it should be understood that term “include” or “have” indicates that a feature, a number, a step, an operation, a component, a part or the combination thereof described in the present disclosure is present, but does not exclude a possibility of presence or addition of one or more other features, numbers, steps, operations, components, parts or combinations thereof, in advance.

The following technology may be used in various radio access system including CDMA, FDMA, TDMA, OFDMA, SC-FDMA, and the like. The CDMA may be implemented as radio technology such as Universal Terrestrial Radio Access (UTRA) or CDMA2000. The TDMA may be implemented as radio technology such as a global system for mobile communications (GSM)/general packet radio service (GPRS)/enhanced data rates for GSM evolution (EDGE). The OFDMA may be implemented as radio technology such as Institute of Electrical and Electronics Engineers (IEEE) 802.11 (Wi-Fi), IEEE 802.16 (WiMAX), IEEE 802.20, Evolved UTRA (E-UTRA), or the like. The UTRA is a part of Universal Mobile Telecommunications System (UMTS). 3rd Generation Partnership Project (3GPP) Long Term Evolution (LTE) is a part of Evolved UMTS (E-UMTS) using the E-UTRA and LTE-Advanced (A)/LTE-A pro is an evolved version of the 3GPP LTE. 3GPP NR (New Radio or New Radio Access Technology) is an evolved version of the 3GPP LTE/LTE-A/LTE-A pro. 3GPP 6G may be an evolved version of 3GPP NR.

For clarity of description, the technical spirit of the present disclosure is described based on the 3GPP communication system (e.g., LTE or NR etc), but the technical spirit of the present disclosure are not limited thereto. LTE means technology after 3GPP TS 36.xxx Release 8. In detail, LTE technology after 3GPP TS 36.xxx Release 10 is referred to as the LTE-A and LTE technology after 3GPP TS 36.xxx Release 13 is referred to as the LTE-A pro. The 3GPP NR means technology after TS 38.xxx Release 15. 3GPP 6G may mean technology after TS Release 17 and/or Release 18. "xxx" means standard document detail number. LTE/NR/6G may be collectively referred to as a 3GPP system. For background art, terms, abbreviations, etc. used in the description of the present disclosure, reference may be made to matters described in standard documents published prior to the present disclosure. For example, the following document can be referred to.

3GPP LTE

- 36.211: Physical channels and modulation
- 36.212: Multiplexing and channel coding
- 36.213: Physical layer procedures for control
- 36.300: Overall description
- 36.331: Radio Resource Control (RRC)

3GPP NR

- 38.211: Physical channels and modulation
- 38.212: Multiplexing and channel coding
- 38.213: Physical layer procedures for control
- 38.214: Physical layer procedures for data
- 38.300: NR and NG-RAN Overall Description
- 38.331: Radio Resource Control (RRC) protocol specification

Physical Channels and Frame Structure

Physical Channel and General Signal Transmission

FIG. 1 illustrates physical channels and general signal transmission used in a 3GPP system. In a wireless communication system, the UE receives information from the eNB through Downlink (DL) and the UE transmits information from the eNB through Uplink (UL). The information which the eNB and the UE transmit and receive includes data and various control information and there are various physical channels according to a type/use of the information which the eNB and the UE transmit and receive.

When the UE is powered on or newly enters a cell, the UE performs an initial cell search operation such as synchronizing with the eNB (S11). To this end, the UE may receive a Primary Synchronization Signal (PSS) and a (Secondary Synchronization Signal (SSS) from the eNB and synchronize with the eNB and acquire information such as a cell ID or the like. Thereafter, the UE may receive a Physical Broadcast Channel (PBCH) from the eNB and acquire in-cell broadcast information. Meanwhile, the UE receives a Downlink Reference Signal (DL RS) in an initial cell search step to check a downlink channel status.

A UE that completes the initial cell search receives a Physical Downlink Control Channel (PDCCH) and a Physical Downlink Control Channel (PDSCH) according to information loaded on the PDCCH to acquire more specific system information (S12).

Meanwhile, when there is no radio resource first accessing the eNB or for signal transmission, the UE may perform a Random Access Procedure (RACH) to the eNB (S13 to S16). To this end, the UE may transmit a specific sequence to a preamble through a Physical Random Access Channel (PRACH) (S13 and S15) and receive a response message (Random Access Response (RAR) message) for the preamble through the PDCCH and a corresponding PDSCH. In

the case of a contention based RACH, a Contention Resolution Procedure may be additionally performed (S16).

The UE that performs the above procedure may then perform PDCCH/PDSCH reception (S17) and Physical Uplink Shared Channel (PUSCH)/Physical Uplink Control Channel (PUCCH) transmission (S18) as a general uplink/downlink signal transmission procedure. In particular, the UE may receive Downlink Control Information (DCI) through the PDCCH. Here, the DCI may include control information such as resource allocation information for the UE and formats may be differently applied according to a use purpose.

Meanwhile, the control information which the UE transmits to the eNB through the uplink or the UE receives from the eNB may include a downlink/uplink ACK/NACK signal, a Channel Quality Indicator (CQI), a Precoding Matrix Index (PMI), a Rank Indicator (RI), and the like. The UE may transmit the control information such as the CQI/PMI/RI, etc., through the PUSCH and/or PUCCH.

Structures of Uplink and Downlink Channels

Downlink Channel Structure

The BS transmits an associated signal to the UE through a downlink channel to be described below and the UE receives the associated signal from the BS through the downlink channel to be described below.

(1) Physical Downlink Shared Channel (PDSCH)

The PDSCH transports downlink data (e.g., DL-shared channel transport block (DL-SCH TB)), and adopts modulation methods such as Quadrature Phase Shift Keying (QPSK), 16 Quadrature Amplitude Modulation (QAM), 64 QAM, and 256 QAM. A codeword is generated by encoding a TB. The PDSCH may carry multiple codewords. Scrambling and modulation mapping are performed for each codeword and modulation symbols generated from each codeword are mapped to one or more layers (layer mapping). Each layer is mapped to a resource together with a demodulation reference signal (DMRS), generated as an OFDM symbol signal, and transmitted through a corresponding antenna port.

(2) Physical Downlink Control Channel (PDCCH)

The PDCCH transports downlink control information (DCI) and a QPSK modulation method is applied. One PDCCH is constituted by 1, 2, 4, 8, and 16 Control Channel Elements (CCEs) according to an Aggregation Level (AL). One CCE is constituted by 6 Resource Element Groups (REGs). One REG is defined by one OFDM symbol and one (P)RB.

The UE performs decoding (so-called, blind decoding) for a set of PDCCH candidates to obtain the DCI transmitted through the PDCCH. The set of PDCCH candidates decoded by the UE is defined as a PDCCH search space set. The search space set may be a common search space or a UE-specific search space. The UE may obtain the DCI by monitoring PDCCH candidates in one or more search space sets configured by the MIB or higher layer signaling.

Uplink Channel Structure

The UE transmits an associated signal to the BS through an uplink channel to be described below and the BS receives the associated signal from the UE through the uplink channel to be described below.

(1) Physical Uplink Shared Channel (PUSCH)

The PUSCH transports uplink data (e.g., UL-shared channel transport block (UL-SCH TB) and/or uplink control information (UCI) and is transmitted based on a Cyclic Prefix-Orthogonal Frequency Division Multiplexing (CP-OFDM) waveform or a Discrete Fourier Transform-spread-Orthogonal Frequency Division Multiplexing (DFT-s-

OFDM) waveform. When the PUSCH is transmitted based on the DFT-s-OFDM waveform, the UE transmits the PUSCH by applying transform precoding. As an example, when the transform precoding is disabled (e.g., transform precoding is disabled), the UE transmits the PUSCH based on the CP-OFDM waveform, and when the transform precoding is enabled (e.g., transform precoding is enabled), the UE may transmit the PUSCH based on the CP-OFDM waveform or the DFT-s-OFDM waveform. PUSCH transmission is dynamically scheduled by the UL grant in the DCI or semi-statically scheduled based on higher layer (e.g., RRC) signaling (and/or Layer 1 (L1) signaling (e.g., PDCCH)) (configured grant). The PUSCH transmission may be performed based on a codebook or a non-codebook.

(2) Physical Uplink Control Channel (PUCCH)

The PUCCH may transport uplink control information, HARQ-ACK, and/or scheduling request (SR), and may be divided into multiple PUCCHs according to a PUCCH transmission length.

General Contents of 6G System

A 6G (wireless communication) system has purposes including (i) a very high data rate per device, (ii) a very large number of connected devices, (iii) global connectivity, (iv) a very low latency, (v) reduction of energy consumption of battery-free IoT devices, (vi) ultrahigh reliability connection, (vii) connected intelligence having a machine learning capability, etc. A vision of the 6G system may be four aspects, e.g., intelligent connectivity, deep connectivity, holographic connectivity, and ubiquitous connectivity, and the 6G system may satisfy requirements shown in Table 1 below. That is, Table 1 shows an example of the requirements of the 6G system.

TABLE 1

Per device peak data rate	1 Tbps
E2E latency	1 ms
Maximum spectral efficiency	100 bps/Hz
Mobility support	Up to 1000 km/hr
Satellite integration	Fully
AI	Fully
Autonomous vehicle	Fully
XR	Fully
Haptic Communication	Fully

The 6G system may have key factors such as Enhanced mobile broadband (eMBB), Ultra-reliable low latency communications (URLLC), massive machine-type communication (mMTC), AI integrated communication, Tactile internet, High throughput, High network capacity, High energy efficiency, Low backhaul and access network congestion, and Enhanced data security. FIG. 2 is a diagram illustrating an example of a communication structure providable in a 6G system.

It is anticipated that the 6G system has higher simultaneous wireless communication connectivity which is 50 times higher than the 5G wireless communication system. The URLLC which is the key feature of 5G will be a more important technology by providing a smaller end-to-end latency than 1 ms in the 6G communication. In the 6G system, volume spectrum efficiency will be even more excellent unlike region spectrum efficiency frequently used. The 6G system may provide an advanced battery technology for a very long battery life-span and energy harvesting. In 6G, new network features may be as follows.

Satellites integrated network: It is anticipated that the 6G will be integrated with the satellite in order to provide a global mobile group. Integrating the terrestrial, the

satellite, and the aerial network into one wireless communication system is very important for the 6G.

Connected intelligence: Unlike the wireless communication system of the previous generation, the 6G is innovative, and wireless evolution will be updated from “connected things” to “connected intelligence”. The AI may be applied in each step (or each procedure of signal processing to be described below) of a communication procedure.

Seamless integration wireless information and energy transfer: The 6G wireless network will transfer power in order to charge batteries of devices like smartphones and sensors. Therefore, wireless information and energy transmission (WIET) will be integrated.

Ubiquitous super 3D connectivity: Connection to networks of networks and core network functions of a drone and a very low earth orbit network will make a super 3D connection in 6G ubiquitous.

Several general requirements may be as follows in new network features of the 6G described above.

Small cell networks: An ID of the small cell network is introduced in order to enhance a received signal quality as a result of enhancement of a throughput, energy efficiency and spectrum efficiency in a cellular system. Consequently, the small cell network is a required feature of communication systems of 6G and beyond 5G (5 GB) or more. Therefore, the 6G communication system also adopts the feature of the small cell network.

Ultra-dense heterogeneous network: Ultra-dense heterogeneous networks will become another important feature of the 6G communication system. A multi-tier network constituted by heterogeneous networks improves entire QoS and reduces cost.

High-capacity backhaul: The backhaul connectivity is characterized as the high-capacity backhaul network in order to support high-capacity traffic. A high-speed optical fiber and free space optic system may be an available solution for such a problem.

Radar technology integrated with mobile technology: High-precision localization (location based service) through communication is one of the functions of the 6G wireless communication system. Therefore, a radar system will be integrated with the 6G network.

Softwarization and virtualization: The softwarization and virtualization are two important functions which become a basis of a design process in the 5 GB network in order to guarantee reconfigurability and programmability. Further, in a sharing physical infrastructure, billions of devices may be shared.

Core Implementation Technology of 6G System Artificial Intelligence

The AI is most important in the 6G system, and is a technology to be newly introduced. The AI is not involved in the 4G system. The 5G system will support a partial or very limited AI. However, the 6G system will completely support the AI for automation. Development of machine learning will make a more intelligent network for real-time communication in the 6G. When the AI is introduced into communication, real-time data transmission may be simplified and enhanced. The AI may determine a scheme in which a complicated target task is performed by using numerous analyses. That is, the AI may increase efficiency and decrease the processing latency.

Time-consuming task such as handover, network selection, and resource scheduling may be immediately performed by using the AI. The AI may play an important role even in M2M, machine-to-human, and human-to-machine

communication. Further, the AI may become rapid communication in a brain computer interface (BCI). An AI based communication system may be supported by a metal material, an intelligent structure, an intelligent network, an intelligent device, an intelligent cognition radio, a self-maintenance network, and machine learning.

In recent years, an attempt to integrate the AI with the wireless communication system has appeared, but this concentrates an application layer and a network layer, in particular, deep learning on a wireless resource management and allocation field. However, this study is gradually developed to a MAC layer and a physical layer, and in particular, there are attempts to combine the deep learning with wireless transmission in a physical layer. AI based physical layer transmission means not a traditional communication framework, but a signal processing and communication mechanism based on an AI driver in a fundamental signal processing and communication mechanism. For example, the AI based physical layer transmission may include channel coding and decoding, deep learning based signal estimation and detection, a deep learning based MIMO mechanism, AI based resource scheduling and allocation, etc.

The machine learning may be used for channel estimation and channel tracking, and used for power allocation, interference cancellation, etc., in a downlink physical layer. Further, the machine learning may also be used for antenna selection, power control, symbol detection, etc., in an MIMO system.

However, application of DNN for transmission in the physical layer may have the following problem.

In a deep learning based AI algorithm, numerous training data are required to optimize training parameters. However, due to a limit in acquiring data in a specific channel environment as the training data, a lot of training data are used offline. This, in the specific channel environment, with respect to static training for the training data, a contradiction may occur between dynamic features and diversity of a radio channel.

Further, current deep learning primarily targets an actual real signal. However, signals of the physical layer of the wireless communication are complex signals. In order to match features of wireless communication signals, a study for a neural network which detects a complex domain signal is further required.

Hereinafter, the machine learning will be described in more detail.

The machine learning means a series of operations of training the machine in order to make a machine which may perform a task which a person may perform or which it is difficult for the person to perform. Data and a learning model are required for the machine learning. A learning method of the data in the machine learning may be generally classified into three, i.e., supervised learning, unsupervised learning, and reinforcement learning.

Learning of a neural network is to minimize an error of an output. The learning of the neural network is a process of repeatedly inputting learning data into the neural network and calculating the output of the neural network for the learning data and the error of a target and back-propagating the errors of the neural network from the output layer of the neural network toward the input layer in a direction to reduce the errors to update the weight of each node of the neural network.

The supervised learning may use learning data in which the learning data is labeled with a correct answer, and the unsupervised learning may use learning data in which not be labeled with the correct answer. That is for example, the

learning data in the case of the supervised learning related to the data classification may be data in which category is labeled in each learning data. The labeled learning data is input to the neural network, and the error may be calculated by comparing the output (category) of the neural network with the label of the learning data. The calculated error is back-propagated in a reverse direction (i.e., a direction from the output layer toward the input layer) in the neural network and connection weights of respective nodes of each layer of the neural network may be updated according to the back propagation. A variation amount of the updated connection weight of each node may be determined according to a learning rate. Calculation of the neural network for the input data and the back-propagation of the error may constitute a learning cycle (epoch). A learning rate may be applied differently according to the number of repetition times of the learning cycle of the neural network. For example, in an initial stage of the learning of the neural network, the neural network ensures a certain level of performance quickly by using a high learning rate, thereby increasing efficiency and a low learning rate is used in a latter stage of the learning, thereby increasing accuracy.

The learning method may vary depending on a feature of data. For example, when a purpose is to accurately predict data by a transmitter by a receiver on the communication system, it is more preferable to perform learning by using the supervised learning than the unsupervised learning or reinforcement learning.

The learning model corresponds to a human brain, and a most basic linear model may be considered, but a paradigm of a machine learning a neural network structure having high complexity such as artificial neural networks as the learning model is referred to the deep learning.

A neural network core used as the learning scheme generally includes deep neural networks (DNN), convolutional deep neural networks (CNN), and recurrent neural networks (RNN) (recurrent Boltzman machine) schemes.

The artificial neural network is an example of connecting several perceptrons.

Referring to FIG. 3, when an input vector $x=(18, 19, \dots, x_d)$ is input, an entire process of multiplying each component by weights (W_1, W_2, W_d), and aggregating all of the results are aggregated, and then applying an active function $\sigma(\cdot)$ is referred to as perceptron. A massive artificial neural network structure extends a simplified perceptron structure illustrated in FIG. 3 to apply the input vector to different multi-dimension perceptrons. For convenience of description, an input value or an output value is referred to as a node.

Meanwhile, it may be described that the perceptron structure illustrated in FIG. 3 is constituted by a total of three layers based on the input value and the output value. An artificial neural network in which there are H $(d+1)$ -dimension perceptrons between a 1st layer and a 2nd layer and K $(H+1)$ -dimension perceptrons between the 2nd layer and a 3rd layer may be expressed as in FIG. 4.

An input layer on which the input vector is located is referred to as an input layer, an output layer on which a final output value is located is referred to as an output layer, and all layers located between the input layer and the output layer are referred to as a hidden layer. In the illustration of FIG. 4, three layers are disclosed, but when the actual number of artificial neural network layers is counted, the number is counted except for the input layer, so the actual number of artificial neural network layers may be regarded

as a total of two layers. The artificial neural network is configured by connecting a perceptron of a basic block in two dimensions.

The input layer, hidden layer, and output layer may be commonly applied in various artificial neural network structures such as CNN, RNN, etc., to be described below in addition to a multi-layer perceptron. As the number of hidden layers increases, the artificial neural network is deeper, and a machine learning paradigm that uses a sufficiently deepened artificial neural network as the learning model is referred to as deep learning. Further, the artificial neural network used for the deep learning is referred to as a deep neural network (DNN).

The DNN illustrated in FIG. 5 is a multi-layer perceptron constituted by eight hidden layer+output layer. The multi-layer perceptron structure is expressed as a fully-connected neural network. In the fully-connected neural network, there is no connection relationship between nodes located on the same layer, and there is the connection relationship only between nodes located on adjacent layers. The DNN has a fully-connected neural network structure, and is configured by a combination of multiple hidden layers and active functions, and may be usefully applied to determining correlation features between an input and an output. Here, the correlation features may mean a joint probability of the input and the output.

Meanwhile, various artificial neural network structures different from the DNN may be formed according to how a plurality of perceptrons being connected to each other.

In the DNN, nodes located inside one layer are disposed in a 1-dimensional vertical direction. However, in FIG. 6, a case where w horizontal nodes and h vertical nodes are disposed in two dimensions with respect to the nodes may be assumed (a convolutional neural network structure of FIG. 6). In this case, since a weight is added per one connection in a connection process connected from one input node to the hidden layer, a total of $h \times w$ weights should be considered. Since there are $h \times w$ nodes in the input layer, a total of $h^2 w^2$ weights are required between two adjacent layers.

The convolutional neural network of FIG. 6 has a problem in that the number of weights exponentially increases according to the number of connections, so by assuming that there is a small-sized filter instead of considering all mode connections between the adjacent layers, a weighted sum and an active function operation are performed with respect to a portion where the filters are overlapped as illustrated in FIG. 7.

One filter has weights corresponding to a number as large as a size thereof, and the weight may be learned so as to extract and output any specific feature on the image as a factor. In FIG. 7, a 3×3 -sized filter is applied to an uppermost left end 3×3 region of the input layer, and an output value of a result of performing the weighted sum and the active function operation for the corresponding node is stored in z_{22} .

The filter performs the weighted sum and active function operation while moving at predetermined horizontal and vertical intervals while scanning the input layer, and the output value thereof is located at a position of a current filter. Such an operation scheme is similar to a conventional operation for an image in a computer vision field, so the DNN having such a structure is referred to as a convolutional neural network (CNN) and a hidden layer generated as the result of the convolutional operation is referred to as a convolutional layer. Further, a neural network in which there is a plurality of convolutional layers is referred to as a deep convolutional neural network (DCNN).

In the convolutional layer, in the node at which the current filter is located, only a node located at a region covered by the filter is included to calculate the weighted sum, thereby reducing the number of weights. As a result, one filter may be used to concentrate on a feature for a local region. As a result, the CNN may be effectively applied to image data processing in which a physical distance in a 2D region becomes an important determination criterion. Meanwhile, in the CNN, a plurality of filters may be applied just before the convolutional layer, and a plurality of output results may also be generated through the convolutional operation of each filter.

Meanwhile, there may be data in which sequence features are important. A structure applying a scheme of inputting one element on a data sequence every timestep by considering a length variability and a time order of the sequence data, and inputting an output vector (hidden vector) of the hidden layer output at a specific time jointly with a just next element on the sequence to the artificial neural network is referred to a recurrent neural network structure.

Referring to FIG. 8, the recurrent neural network (RNN) is a structure of applying the weighted sum and the active function by jointly inputting hidden vectors ($z_1(t-1)$, $z_2(t-1)$, $\dots$, $z_H(t-1)$) at an immediately previous time $t-1$ in the process of elements ($18(t)$, $19(t)$, $\dots$, $xd(t)$) at any time t on the data sequence. A reason for transferring the hidden vector to a subsequent time is that it is regarded that information in the input vector at the previous times is accumulated in the hidden vector at the current time.

Referring to FIG. 8, the RNN operates according to a predetermined time order with respect to the input data sequence.

A hidden vector ($z_1(1)$, $z_2(1)$, $\dots$, $z_H(1)$) when an input vector ($18(1)$, $19(1)$, $\dots$, $xd(1)$) at time 1 is input into the RNN is input jointly with an input vector ($18(2)$, $19(2)$, $\dots$, $xd(2)$) at time 2 to determine a vector ($z_1(2)$, $z_2(2)$, $\dots$, $z_H(2)$) of the hidden layer through the weighted sum and the active function. Such a process is repeatedly performed up to time 2, time 3, $\dots$, time T .

Meanwhile, when a plurality of hidden layers is disposed in the RNN, this is referred to as a deep recurrent neural network (DRNN). The RNN is usefully applied to the sequence data (e.g., natural language processing) and designed.

The neural network core used as the learning scheme may include various deep learning techniques such as a Restricted Boltzmann Machine (RBM), deep belief networks (DBN), and a Deep Q-Network in addition to the DNN, the CNN, and the RNN, and may be applied to fields such as computer vision, voice recognition, natural language processing, voice/signal processing, etc.

In recent years, an attempt to integrate the AI with the wireless communication system has appeared, but this concentrates an application layer and a network layer, in particular, deep learning on a wireless resource management and allocation field. However, this study is gradually developed to a MAC layer and a physical layer, and in particular, there are attempts to combine the deep learning with wireless transmission in a physical layer. AI based physical layer transmission means not a traditional communication framework, but a signal processing and communication mechanism based on an AI driver in a fundamental signal processing and communication mechanism. For example, the AI based physical layer transmission may include channel coding and decoding, deep learning based signal estimation and detection, a deep learning based MIMO mechanism, AI based resource scheduling and allocation, etc.

Terahertz (THz) Communication

The data rate may increase a bandwidth. This may be performed by using sub-THz communication with a wide bandwidth, and applying an advanced massive MIMO technology. A THz wave also known as a radiation of millimeters or less generally shows a frequency band between 0.1 THz and 10 THz having a corresponding wavelength in the range of 0.03 mm to 3 mm. A 100 GHz to 300 GHz band range (sub THz band) is regarded as a primary part of the THz for cellular communication. When the sub-THz band is added to a mmWave band, a 6G cellular communication capacity increases. 300 GHz to 3 THz in a defined THz band is present in an infrared (IR) frequency band. The 300G Hz to 3 THz band is a part of a wideband, but is present on a boundary of the wideband, and is present just behind an RF band. Therefore, the 300G Hz to 3 THz band shows a similarity to the RF. FIG. 10 illustrates an example of an electromagnetic spectrum.

The main characteristics of THz communications include (i) widely available bandwidth to support very high data rates, and (ii) high path loss at high frequencies (highly directional antennas are indispensable). The narrow beamwidth produced by the highly directional antenna reduces interference. The small wavelength of the THz signal allows a much larger number of antenna elements to be incorporated into devices and BSs operating in this band. This enables advanced adaptive array technology to overcome range limitations.

Optical Wireless Technology

An OWC technology is planned for 6G communication in addition to RF based communication for all available device-to-access networks. The network is connected to a network-to-backhaul/fronthaul network connection. The OWC technology is already used after the 4G communication system, but may be more widely used for meeting the requirements of the 6G communication system. The OWC technologies such as light fidelity, visible light communication, optical commerce communication, and FSO communication based on the wideband are already well known technologies. The optical wireless technology based communication may provide a very high data speed, a low latency time, and safe communication. LiDAR may also be used for superultra high resolution 3D mapping in the 6G communication based on the wideband.

FSO Backhaul Network

Transmitter and receiver features of the FSO system are similar to the features of the optical fiber network. Therefore, data transmission of the FSO system is similar to that of the optical fiber system. Therefore, the FSO may become an excellent technology that provides the backhaul connection in the 6G system jointly with the optical fiber network. When the FSO is used, very long-distance communication is enabled even in a distance of 10000 km or more. The FSO supports a massive backhaul connection for remote and non-remote regions such as the sea, the space, the underwater, and an isolated island. The FSO also supports a cellular BS connection.

Massive MIMO Technology

One of the core technologies for enhancing the spectrum efficiency is application of the MIMO technology. When the MIMO technology is enhanced, the spectrum efficiency is also enhanced. Therefore, the massive MIMO technology will be important in the 6G system. Since the MIMO technology uses multiple paths, a multiplexing technology and a beam generation and operating technology suitable for the THz band should also be considered importantly so that the data signal is transmitted through one or more paths.

Blockchain

The blockchain will become an important technology for managing mass data in a future communication system. The blockchain is a form of a distributed ledger technology, and the distributed ledger is a database distributed in numerous nodes or computing devices. Each node replicates and stores the same ledger copy. The blockchain is managed by a P2P network. The blockchain may be present without being managed by a centralized agency or server, but may be present. Data of the blockchain is jointly collected and constituted by blocks. The blocks are connected to each other, and protected by using encryption. The blockchain fundamentally perfectly complements massive IoT through enhanced interoperability security, personal information protection, stability, and scalability. Therefore, the blockchain technology provides various functions such as inter-device interoperability, massive data traceability, autonomous interactions of different IoT systems, and massive connection stability of the 6G communication system.

3D Networking

The 6G system supports user communication of vertical scalability by integrating ground and aerial networks. A 3D BS will be provided through a low-orbit satellite and a UAV. When a new dimension is added in terms of an altitude and a related degree of freedom, a 3D connection is quite different from the existing 2D network.

Quantum Communication

In the context of the 6G network, unsupervised reinforcement learning of the network is promising. The supervised learning scheme may designate a label in a vast amount of data generated in the 6G. The labeling is not required in the unsupervised learning. Therefore, the technology may be used for autonomously constructing a complicated network expression. When the reinforcement learning and the unsupervised learning are combined, the network may be operated by a true autonomous scheme.

Unmanned Aerial Vehicle (UAV)

The unmanned aerial vehicle (UAV) or a drone will become an important element in the 6G wireless communication. In most cases, a high-speed data wireless connection is provided by using a UAV technology. ABS entity is installed in the UAV in order to provide the cellular connection. The UAV has a specific function which may not be seen in a fixed BS infrastructure, such as easy deployment, strong visible-line link, the degree of freedom in which mobility is controlled. During emergency situations such as natural disaster, the placement of a ground communication infrastructure is not enabled to be economically realized, and sometimes may not provide services in volatile environments. The UAV may easily handle this situation. The UAV will become a new paradigm of a wireless communication field. This technology facilitates three basic requirements of the wireless network, i.e., eMBB, URLLC, and mMTC/The UAV may also support various purposes such as network connectivity enhancement, fire sensing, disaster emergency services, securing and monitoring, pollution monitoring, parking monitoring, accident monitoring, etc. Therefore, the UAV technology is recognized as one of the most important technologies for the 6G communication.

Cell-Free Communication

The close integration of multiple frequencies and heterogeneous communication technologies is very important in the 6G system. As a result, the user may move smoothly from the network to another network without the need of creating any manual configuration in the device. A best network is automatically selected in an available communication technology. This will break the limitation of a cell

concept in the wireless communication. Currently, user movement from one cell to another cell causes too many handovers in the network, and causes handover failures, handover latency, data loss, and pingpong effects. 6G cell-free communication will overcome all of the problems, and provide a better QoS. The cell-free communication will be achieved through different heterogeneous radios of multi-connectivity and multi-tier hybrid technologies and devices. Wireless Information and Energy Transmission Integration

WIET uses the same field and wave as the wireless communication system. In particular, the sensor and the smartphone will be charged by using wireless power transmission during communication. The WIET is a promising technology for extending the life-span of a battery charging wireless system. Therefore, a device without the battery will be supported in the 6G communication.

Integration of Sensing and Communication

An autonomous wireless network is a function to continuously an environmental state which is dynamically changed, and exchange information between different nodes. In the 6G, the sensing will be closely integrated with the communication in order to an autonomous system.

Integration of Access Backhaul Network

In the 6G, the density of the access network will be enormous. Each access network is connected by the backhaul network such as the optical fiber and the FSO network. In order to cope with a very larger number of access networks, there will be a close integration between the access and the backhaul network.

Hologram and Beamforming

The beamforming is a signal processing procedure of adjusting an antenna array in order to transmit a radio signal. The beamforming is a sub-set of a smart antenna or advanced antenna system. The beamforming technology has several advantages such as a high call-to-noise ratio, interference prevention and denial, and high network efficiency. The hologram and beamforming (HBF) is a new beamforming method which is significantly different from the MIMO system because a software-defined antenna is used. The HBF will be a very effective approach scheme for efficient and flexible transmission and reception of the signal in a multi-antenna communication device.

Big Data Analysis

The big data analysis is a complicated process for analyzing various large-scale data sets or big data. This process guarantees perfect data management by finding hidden data, and information such as a correlation and a customer tendency which may not be known. The big data is collected from various sources such as a video, a social network, an image, and the sensor. This technology is widely used to processing vast data in the 6G system.

Large Intelligent Surface (LIS)

A THz band signal has a strong straightness, so there may be a lot of shade regions due to obstacles, and an LIS technology will be important in which the LIS is installed near such a shade region to expand a communication zone and to enable communication stability strengthening and additional services. The LIS is an artificial surface made of electromagnetic materials, and may change propagations of incoming radio waves and outgoing radio waves. The LIS may be shown as an extension of massive MIMO, but is different from the massive MIMO in terms of an array structure and an operating mechanism. Further, the LIS has an advantage of maintaining low power consumption in that the LIS operates as a reconfigurable reflector having passive element, i.e., reflects the signal only passively without using an active RF chain. Further, since each passive reflector of

the LIS should independently control a phase shift of an incident signal, the reflector may be advantage for the wireless communication channel. By appropriately controlling the phase shift through an LIS controller, a reflected signal may be gathered in a target receiver in order to boost a received signal power.

Overview of Terahertz (THz) Wireless Communication

THz wireless communication uses wireless communication using THz waves having a frequency of approximately 0.1 to 10 THz (1 THz=1012 Hz), and may mean wireless communication in a terahertz (THz) band using a very high carrier frequency of 100 GHz or more. THz waves are located between a radio frequency (RF)/millimeter (mm) band and an infrared band, and (i) is transmitted through a non-metal/non-polarizable material better than visible light/infrared rays, and have high straightness due to a shorter wavelength than that of the RF/millimeter waves, and beam focusing may be possible. In addition, since photon energy of the THz waves is only a few meV, the THz waves is harmless to human bodies. A frequency band expected to be used for THz wireless communication may be a D-band (110 GHz to 170 GHz) or an H-band (220 GHz to 325 GHz) band with a low propagation loss due to molecular absorption in the air. Standardization of the THz wireless communication is being discussed mainly in the IEEE 802.15 THz working group, in addition to 3GPP, and content described in the present disclosure can be specified or supplemented by standard documents issued by Task Group (TG3d and TG3e) of the IEEE 802.15. The THz wireless communication may be applied to wireless cognition, sensing, imaging, wireless communication, THz navigation, and the like.

FIG. 11 is a diagram illustrating an example of a THz communication application.

As illustrated in FIG. 11, a THz wireless communication scenario can be classified into a macro network, a micro network, and a nanoscale network. In the macro network, the THz wireless communication may be applied to a vehicle-to-vehicle connection and a backhaul/fronthaul connection. In the micro network, the THz wireless communication may be applied to an indoor small cell, a fixed point-to-point or multi-point connection such as a wireless connection in a data center, and near-field communication such as kiosk downloading.

Table 2 below is a table illustrating an example of a technology that can be used in THz waves.

TABLE 2

Transceivers Device	Available immature: UTC-PD, RTD and SBD
Modulation and coding	Low order modulation techniques (OOK, QPSK), LDPC, Reed Soloman, Hamming, Polar, Turbo
Antenna	Omni and Directional, phased array with low number of antenna elements
Bandwidth	69 GHz (or 23 GHz) at 300 GHz
Channel models	Partially
Data rate	100 Gbps
Outdoor deployment	No
Free space loss	High
Coverage	Low
Radio Measurements	300 GHz indoor
Device size	Few micrometers

The THz wireless communication can be classified according to a method of generating and receiving THz. A THz generation method can be classified as an optical element or electronic element-based technology. FIG. 12 is a diagram illustrating an example of an electronic element-based THz wireless communication transceiver.

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Examples of a method of generating THz using an electronic device include a method using a semiconductor device such as a resonant tunneling diode (RTD), a method using a local oscillator and a multiplier, and a monolithic microwave integrated circuit (MMIC) method using a compound semiconductor high electron mobility transistor (HEMT)-based integrated circuit, and a method using a Si-CMOS-based integrated circuit. In the case of FIG. 12, a multiplier (a doubler or a tripler) is applied to increase a frequency, and a signal passes through a subharmonic mixer and is emitted by an antenna. Since a THz band forms a high frequency, a multiplier is essential. Here, the multiplier is a circuit that provides an output frequency that is N times the frequency of an input, and the multiplier performs matching with a desired harmonic frequency, and filters all other frequencies. For the antenna of FIG. 12, an array antenna or the like may be applied to implement beamforming. In FIG. 12, IF denotes an intermediate frequency, a tripler and a multiplier denote a multiplier, PA denotes a power amplifier, LNA denotes a low noise amplifier, and PLL denotes a phase-locked loop.

FIG. 13 is a diagram illustrating an example of a method of generating a THz signal based on an optical element, and FIG. 14 is a diagram illustrating an example of a THz wireless communication transceiver based on an optical element.

An optical element-based THz wireless communication technology refers to a method of generating and modulating a THz signal using an optical element. An optical element-based THz signal generation technology is a technology for generating an ultra-high speed optical signal using a laser and an optical modulator, and converting the optical signal into a THz signal using an ultra-high speed photodetector. With this technology, it is easy to increase a frequency, and it is possible to generate a high-power signal and obtain a flat response characteristic in a wide frequency band, as compared to a technology using only an electronic device. For optical element-based THz signal generation, as illustrated in FIG. 13, a laser diode, a broadband optical modulator, and a high-speed photodetector are required. In the case of FIG. 13, two laser light signals having different wavelengths are combined to generate a THz signal corresponding to a difference in wavelength between two lasers. In FIG. 13, an optical coupler is a semiconductor device that transmits an electrical signal using light waves in order to provide coupling to electrical isolation between circuits or systems, and a Uni-Travelling Carrier Photo-Detector (UTC-PD) is one photodetector, and is a device in which electrons are used as active carriers and a movement time of the electrons is reduced through bandgap grading. A UTC-PD is capable of photodetection at 150 GHz or more. In FIG. 14, an erbium-doped fiber amplifier (EDFA) is an erbium-added optical fiber amplifier, a photodetector (PD) is a semiconductor device capable of converting an optical signal into an electrical signal, an OSA is an optical module (Optical Sub Assembly) obtained by modularizing various optical communication functions (photoelectric conversion, optical to electrical conversion, or the like) into one component, and a DSO is a digital storage oscilloscope.

A structure of an O/E converter will be described with reference to FIGS. 15 and 16. FIG. 15 illustrates a structure of a photonic source-based transmitter, and FIG. 16 illustrates a structure of an optical modulator.

In general, it is possible to change a phase or the like of a signal by causing an optical source of a laser to pass through an optical wave guide. In this case, data is loaded by changing electrical characteristics through a microwave

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contact or the like. Accordingly, an output of the optical modulator is formed as a modulated waveform. A photoelectric modulator (O/E converter) may generate a THz pulse according to an optical rectification operation using a nonlinear crystal, O/E conversion using a photoconductive antenna, emission from a bundle of relativistic electrons, or the like. The THz pulse generated in a manner as described above may have a length in units of femto second to pico second. The O/E converter uses non-linearity of a device to perform down-conversion.

Considering terahertz spectrum usage (THz spectrum usage), several contiguous GHz bands are highly likely to be used as fixed or mobile service usage for a terahertz system. According to outdoor scenario criteria, an available bandwidth may be classified based on oxygen attenuation of 10⁻² dB/km in a spectrum up to 1 THz. Accordingly, a framework in which the available bandwidth consists of several band chunks may be considered. As an example of the framework, when a length of a THz pulse is set to 50 ps for one carrier, a bandwidth (BW) becomes about 20 GHz.

Effective down-conversion from an infrared (IR) band to a THz band depends on how to utilize the non-linearity of the O/E converter. That is, for down-conversion to a desired terahertz band (THz band), a design of an O/E converter having the most ideal non-linearity for a movement to the terahertz band (THz band) is required. When an O/E converter not suitable for a target frequency band is used, an error is highly likely to occur with respect to an amplitude and phase of a pulse.

In a single carrier system, a terahertz transmission and reception system may be implemented using one O/E converter. In some channel environments, as many O/E converters as the number of carriers may be required in a multi-carrier system. In particular, in the case of a multi-carrier system using several broadbands according to a plan related to the above-described spectrum usage, such a phenomenon will be conspicuous. In this regard, a frame structure for the multi-carrier system may be considered. A signal subjected to down-frequency conversion based on the O/E converter may be transmitted in a specific resource area (for example, a specific frame). A frequency domain in the specific resource area may include a plurality of chunks. Each chunk may consist of at least one component carrier (CC).

OFDMA-Based Federated Learning

FIG. 17 is a diagram schematically illustrating OFDMA-based federated learning.

Federated learning is one distributed machine learning scheme, in which several devices (preferably, terminals), which are subjects of learning, share parameters such as a weight or gradient of a local model with a server, and the server collects the parameters of the local models of the respective devices (so-called local parameters) and updates parameters of a global model (so-called global parameters). In this process, since raw data of each device is not shared, it is possible to reduce a communication overhead in a data transmission process and protect personal information.

Existing federated learning based on OFDMA operates as illustrated in FIG. 17. The device transmits a local parameter to each allocated resource, and the server performs offline aggregation on the parameter received from the device. In general, the server derives a global parameter through averaging of all the local parameters and transmits the global parameter to the device again. However, a time taken to update the global parameter is delayed under limited resources as the number of devices participating in learning increases.

AriComp-Based Federated Learning

FIG. 18 is a diagram schematically illustrating AirComp-based federated learning.

To solve this problem, research on AirComp-based federated learning is being conducted in recent years. AirComp is a scheme in which all devices transmit local parameters using the same resources as in FIG. 18, and in a case of a signal received by the server, a sum of local parameters can be naturally obtained due to characteristics of overlap of analog waveforms. Since, in the AirComp-based federated learning, the local parameters are transmitted through the same resources, latency is not greatly influenced by the number of devices participating in the learning. However, the received signal distribution becomes wider as the number of devices participating in learning increases. This is because a probability that a magnitude of an aggregated signal will increase as the number of aggregated received signals increases becomes higher. Therefore, in a situation in which a range of magnitude of a current received signal is limited, it is necessary to reduce the number of devices participating in learning, or to reduce transmission power since a large number of devices participate in learning, to perform transmission. In the former case, a batch size is reduced, and in the latter case, the reliability is reduced, which adversely affects a learning delay time. Therefore, a transmission and reception scheme capable of receiving an aggregated signal at a receiving stage while maintaining the number of devices participating in learning and the transmission power is dealt with in the present disclosure. Meanwhile, in the present disclosure, the device may also be referred to as a terminal.

EXAMPLES

FIG. 19 is a sequence diagram of a method of aligning gradient signs according to an embodiment of the present disclosure.

A scheme for enabling transmission and reception without adjustment of power or the number of terminals participating in learning in the AirComp-based federated learning in a situation in which a range of magnitude of a received signal is limited in the present disclosure will be described.

Prior to dealing with a transmission and reception scheme that will be described, it is assumed that terminals participating in learning are grouped based on CSI information, and signals transmitted by the terminals belonging to the same group have similar reception power sensitivities. In this case, a transmission and reception scheme of the terminals in each group will be described.

In the AirComp-based federated learning, enabling transmission and reception without a loss of power or a loss of the number of devices participating in learning can be seen as intuitively narrowing the received signal distribution. In order to narrow the received signal distribution, various embodiments of the present disclosure consider the following two primary characteristics.

Aggregating as few signals as possible is advantageous in narrowing the aggregated signal distribution.

Aggregating signals having the same direction (for example, sign), if possible, will be helpful to narrow the aggregated signal distribution.

Hereinafter, a method of aligning gradient signs according to various embodiments of the present disclosure will be described with reference to FIG. 19.

Repetition and vector-wise sign clipping are performed on an existing gradient flatten vector, and bias insertion is applied to generate a flatten vector subjected to gradient sign alignment with bias (GSAB).

The flatten vector for an existing gradient having a length N of a k-th terminal in a t-th iteration may be referred to as $\nabla_k^{(t)}$, and the GSAB flatten vector may be referred to as $\nabla_k^{(t)}$.

In an embodiment, the terminal may repeat gradient values included in a first flatten vector and apply clipping and bias values to obtain a second flatten vector having gradient values in which signs have been aligned (S1901). Thereafter, the terminal may transmit the second flatten vector together with the channel information to the server in the form of AirComp (S1902).

In this case, the GSAB flatten vector may be transformed in the form of Equation 1.

$$\nabla_k^{(t)} \rightarrow \nabla_k^{(t)} = [(\nabla_k^{(t)})_+, -b, (\nabla_k^{(t)})_-, +b] \quad [\text{Equation 1}]$$

Here, b is a bias value, which is determined deterministically in advance by the number W of terminals participating in learning and statistical characteristics of the gradients ($\{\nabla_k^{(t)}\}$) of the flatten vector, and thus, a transmission and reception unit (so-called transceiver) uses values defined as a table and promised in advance. According to Equation 1, the signs include positive signs and negative signs, and a negative bias may be applied to values having a positive sign among the gradient values included in the second flatten vector, and a positive bias may be applied to values having a negative sign.

Thereafter, each terminal acquires channel information ($h_k^{(t)}$) in advance, compensates for a phase, and performs transmission in the form of AirComp with the transmission power P, such as a predetermined transmission power. A signal transmitted in this way may be referred to as a gradient signal.

Meanwhile, the server may receive gradient signals for federated learning transmitted in the form of AirComp from a plurality of terminals. Here, the signal may include the channel information and the second flatten vector (see S1902).

Thereafter, the server may obtain an aggregated gradient based on the received gradient signals (S1903) and update parameters of the global model based on the aggregated gradient (S1904). The server may also be hereinafter referred to as a main server.

In this case, the server can receive in the form of Equation 2.

$$r^{(t)} = \sum_{k=1}^W |h_k^{(t)}| \sqrt{P} (\nabla_k^{(t)}) + n \quad [\text{Equation 2}]$$

Here, n means additive white Gaussian noise. The main server acquires an aggregated gradient $\nabla^{(t)}$ of a group through post-processing of the received GSAB sequence as shown in Equation 3 below.

$$\nabla^{(t)} = \frac{r_1^{(t)} + r_2^{(t)}}{P \sum_{k=1}^W |h_k^{(t)}|} \quad [\text{Equation 3}]$$

Here $r_i^{(t)} = [r^{(t)}[n]]_{n=(i-1)N+1}^{iN}$, and $r^{(t)}[n]$ is an n-th entry of $r^{(t)}$. A first part ($r_1^{(t)}$) and a second part ($r_2^{(t)}$) are naturally

offset during an aggregation process due to application of a bias (b) in opposite directions.

Although a single group has been assumed and a transmission and reception method within the group has been described in the present disclosure, the same can apply to a plurality of groups, preferably, a plurality of groups classified according to different CSIs. It is possible to acquire a global gradient through a weighted sum process of the aggregated gradients acquired in the respective groups at a ratio of the number of devices in each group to a total number of devices participating in learning. For example, when the total number of devices participating in learning is W , the number of devices participating in learning in an i -th group is W_i , and the aggregated gradient is $\nabla_i^{(t)}$, a global gradient ($\nabla^{(t)}$) can be acquired in the form of Equation 4.

$$\nabla^{(t)} = \sum_{i=1}^G \frac{W_i}{W} \nabla_i^{(t)} \quad [\text{Equation 4}]$$

Here, resource utilization for repetition may be selectively used according to a situation among time or frequency resources. FIG. 20 is a diagram simply illustrating a resource management method of a proposed scheme.

Experimental Example

Effect of the disclosure proposed in the present disclosure will be described. It is assumed that the gradients obtained by the respective terminals are independently and identically distributed (iid) and have a uniform distribution between $[-1, 1]$, and it is assumed that all channels between the respective terminals and the server have the same size, and a transmission unit performs transmission after compensating for a phase through precoding.

A traditional scheme for performing simple power control in which all terminals participate in learning, a half-and-half scheme (H-H) for performing power control through orthogonal resource utilization in half of all terminals, and a scheme for performing power control using the proposed GSAB are specified.

FIG. 21 is a diagram illustrating a received signal distribution when transmission is performed with the same transmission power P in all of a traditional scheme, an H-H scheme, and a GSAB scheme. As mentioned above, it can be seen that the proposed GSAB scheme has the narrowest distribution.

FIG. 22 is a diagram illustrating a performance comparison between respective schemes when it is assumed that a range of aggregate received signals is $[-1, 1]$. It can be seen that, although the H-H method uses a higher transmission power than the traditional scheme, performance is rather degraded due to an increase in noise. On the other hand, it can be seen that the proposed scheme has about half the MSE value as compared to the traditional scheme. This is because the benefit of an increase in transmission power due to an improved received signal distribution is greater as compared to a loss of reliability due to an increase in noise.

Devices Used in Wireless Communication Systems

Although not limited thereto, various proposals of the present disclosure described above can be applied to various fields requiring wireless communication/connection (eg, 6G) between devices.

Hereinafter, it will be more specifically illustrated with reference to the drawings. In the following drawings/description, the same reference numerals may represent the

same or corresponding hardware blocks, software blocks or functional blocks unless otherwise specified.

FIG. 23 illustrates a communication system 1 applied to the present disclosure.

Referring to FIG. 23, a communication system 1 applied to the present disclosure includes wireless devices, Base Stations (BSs), and a network. Herein, the wireless devices represent devices performing communication using Radio Access Technology (RAT) (e.g., 5G New RAT (NR)) or Long-Term Evolution (LTE)) and may be referred to as communication/radio/5G devices. The wireless devices may include, without being limited to, a robot 100a, vehicles 100b-1 and 100b-2, an eXtended Reality (XR) device 100c, a hand-held device 100d, a home appliance 100e, an Internet of Things (IoT) device 100f, and an Artificial Intelligence (AI) device/server 400. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous driving vehicle, and a vehicle capable of performing communication between vehicles. Herein, the vehicles may include an Unmanned Aerial Vehicle (UAV) (e.g., a drone). The XR device may include an Augmented Reality (AR)/Virtual Reality (VR)/Mixed Reality (MR) device and may be implemented in the form of a Head-Mounted Device (HMD), a Head-Up Display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), and a computer (e.g., a notebook). The home appliance may include a TV, a refrigerator, and a washing machine. The IoT device may include a sensor and a smartmeter. For example, the BSs and the network may be implemented as wireless devices and a specific wireless device 200a may operate as a BS/network node with respect to other wireless devices.

The wireless devices 100a to 100f may be connected to the network 300 via the BSs 200. An AI technology may be applied to the wireless devices 100a to 100f and the wireless devices 100a to 100f may be connected to the AI server 400 via the network 300. The network 300 may be configured using a 3G network, a 4G (e.g., LTE) network, or a 5G (e.g., NR) network. Although the wireless devices 100a to 100f may communicate with each other through the BSs 200/network 300, the wireless devices 100a to 100f may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs/network. For example, the vehicles 100b-1 and 100b-2 may perform direct communication (e.g. Vehicle-to-Vehicle (V2V)/Vehicle-to-everything (V2X) communication). The IoT device (e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices 100a to 100f.

Wireless communication/connection 150a, 150b may be performed between the wireless devices 100a to 100f/base station 200-base station 200/wireless devices 100a to 100f. Here, wireless communication/connection may be performed through various radio access technologies (eg, 5G NR) for uplink/downlink communication 150a and sidelink communication 150b (or D2D communication). Through the wireless communication/connection 150a and 150b, the wireless device and the base station/wireless device may transmit/receive radio signals to each other. For example, the wireless communication/connection 150a and 150b may transmit/receive signals through various physical channels based on all/partial processes of FIG. To this end, based on the various proposals of the present disclosure, various configuration information setting processes for transmitting/

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receiving radio signals, various signal processing processes (eg, channel encoding/decoding, modulation/demodulation, resource mapping/demapping, etc.) At least a part of a resource allocation process may be performed.

FIG. 24 illustrates wireless devices applicable to the present disclosure.

Referring to FIG. 24, a first wireless device 100 and a second wireless device 200 may transmit radio signals through a variety of RATs (e.g., LTE and NR). Herein, {the first wireless device 100 and the second wireless device 200} may correspond to {the wireless device 100x and the BS 200} and/or {the wireless device 100x and the wireless device 100x} of FIG. 23.

The first wireless device 100 may include one or more processors 102 and one or more memories 104 and additionally further include one or more transceivers 106 and/or one or more antennas 108. The processor(s) 102 may control the memory(s) 104 and/or the transceiver(s) 106 and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) 102 may process information within the memory(s) 104 to generate first information/signals and then transmit radio signals including the first information/signals through the transceiver(s) 106. The processor(s) 102 may receive radio signals including second information/signals through the transceiver 106 and then store information obtained by processing the second information/signals in the memory(s) 104. The memory(s) 104 may be connected to the processor(s) 102 and may store a variety of information related to operations of the processor(s) 102. For example, the memory(s) 104 may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) 102 or for performing the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) 102 and the memory(s) 104 may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) 106 may be connected to the processor(s) 102 and transmit and/or receive radio signals through one or more antennas 108. Each of the transceiver(s) 106 may include a transmitter and/or a receiver. The transceiver(s) 106 may be interchangeably used with Radio Frequency (RF) unit(s). In the present disclosure, the wireless device may represent a communication modem/circuit/chip.

The second wireless device 200 may include one or more processors 202 and one or more memories 204 and additionally further include one or more transceivers 206 and/or one or more antennas 208. The processor(s) 202 may control the memory(s) 204 and/or the transceiver(s) 206 and may be configured to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. For example, the processor(s) 202 may process information within the memory(s) 204 to generate third information/signals and then transmit radio signals including the third information/signals through the transceiver(s) 206. The processor(s) 202 may receive radio signals including fourth information/signals through the transceiver(s) 106 and then store information obtained by processing the fourth information/signals in the memory(s) 204. The memory(s) 204 may be connected to the processor(s) 202 and may store a variety of information related to operations of the processor(s) 202. For example, the memory(s) 204 may store software code including commands for performing a part or the entirety of processes controlled by the processor(s) 202 or for performing the

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descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. Herein, the processor(s) 202 and the memory(s) 204 may be a part of a communication modem/circuit/chip designed to implement RAT (e.g., LTE or NR). The transceiver(s) 206 may be connected to the processor(s) 202 and transmit and/or receive radio signals through one or more antennas 208. Each of the transceiver(s) 206 may include a transmitter and/or a receiver. The transceiver(s) 206 may be interchangeably used with RF unit(s). In the present disclosure, the wireless device may represent a communication modem/circuit/chip.

Hereinafter, hardware elements of the wireless devices 100 and 200 will be described more specifically. One or more protocol layers may be implemented by, without being limited to, one or more processors 102 and 202. For example, the one or more processors 102 and 202 may implement one or more layers (e.g., functional layers such as PHY, MAC, RLC, PDCP, RRC, and SDAP). The one or more processors 102 and 202 may generate one or more Protocol Data Units (PDUs) and/or one or more Service Data Unit (SDUs) according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors 102 and 202 may generate messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document. The one or more processors 102 and 202 may generate signals (e.g., baseband signals) including PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document and provide the generated signals to the one or more transceivers 106 and 206. The one or more processors 102 and 202 may receive the signals (e.g., baseband signals) from the one or more transceivers 106 and 206 and acquire the PDUs, SDUs, messages, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document.

The one or more processors 102 and 202 may be referred to as controllers, microcontrollers, microprocessors, or microcomputers. The one or more processors 102 and 202 may be implemented by hardware, firmware, software, or a combination thereof. As an example, one or more Application Specific Integrated Circuits (ASICs), one or more Digital Signal Processors (DSPs), one or more Digital Signal Processing Devices (DSPDs), one or more Programmable Logic Devices (PLDs), or one or more Field Programmable Gate Arrays (FPGAs) may be included in the one or more processors 102 and 202. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software and the firmware or software may be configured to include the modules, procedures, or functions. Firmware or software configured to perform the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be included in the one or more processors 102 and 202 or stored in the one or more memories 104 and 204 so as to be driven by the one or more processors 102 and 202. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document may be implemented using firmware or software in the form of code, commands, and/or a set of commands.

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The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** and store various types of data, signals, messages, information, programs, code, instructions, and/or commands. The one or more memories **104** and **204** may be configured by Read-Only Memories (ROMs), Random Access Memories (RAMs), Electrically Erasable Programmable Read-Only Memories (EPROMs), flash memories, hard drives, registers, cash memories, computer-readable storage media, and/or combinations thereof. The one or more memories **104** and **204** may be located at the interior and/or exterior of the one or more processors **102** and **202**. The one or more memories **104** and **204** may be connected to the one or more processors **102** and **202** through various technologies such as wired or wireless connection.

The one or more transceivers **106** and **206** may transmit user data, control information, and/or radio signals/channels, mentioned in the methods and/or operational flowcharts of this document, to one or more other devices. The one or more transceivers **106** and **206** may receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, from one or more other devices. For example, the one or more transceivers **106** and **206** may be connected to the one or more processors **102** and **202** and transmit and receive radio signals. For example, the one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may transmit user data, control information, or radio signals to one or more other devices. The one or more processors **102** and **202** may perform control so that the one or more transceivers **106** and **206** may receive user data, control information, or radio signals from one or more other devices. The one or more transceivers **106** and **206** may be connected to the one or more antennas **108** and **208** and the one or more transceivers **106** and **206** may be configured to transmit and receive user data, control information, and/or radio signals/channels, mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed in this document, through the one or more antennas **108** and **208**. In this document, the one or more antennas may be a plurality of physical antennas or a plurality of logical antennas (e.g., antenna ports). The one or more transceivers **106** and **206** may convert received radio signals/channels etc. from RF band signals into baseband signals in order to process received user data, control information, radio signals/channels, etc. using the one or more processors **102** and **202**. The one or more transceivers **106** and **206** may convert the user data, control information, radio signals/channels, etc. processed using the one or more processors **102** and **202** from the base band signals into the RF band signals. To this end, the one or more transceivers **106** and **206** may include (analog) oscillators and/or filters.

FIG. 25 illustrates a signal process circuit for a transmission signal.

Referring to FIG. 25, a signal processing circuit **1000** may include scramblers **1010**, modulators **1020**, a layer mapper **1030**, a precoder **1040**, resource mappers **1050**, and signal generators **1060**. An operation/function of FIG. 25 may be performed, without being limited to, the processors **102** and **202** and/or the transceivers **106** and **206** of FIG. 24. Hardware elements of FIG. 25 may be implemented by the processors **102** and **202** and/or the transceivers **106** and **206** of FIG. 24. For example, blocks **1010** to **1060** may be implemented by the processors **102** and **202** of FIG. 24. Alternatively, the blocks **1010** to **1050** may be implemented

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by the processors **102** and **202** of FIG. 24 and the block **1060** may be implemented by the transceivers **106** and **206** of FIG. 24.

Codewords may be converted into radio signals via the signal processing circuit **1000** of FIG. 25. Herein, the codewords are encoded bit sequences of information blocks. The information blocks may include transport blocks (e.g., a UL-SCH transport block, a DL-SCH transport block). The radio signals may be transmitted through various physical channels (e.g., a PUSCH and a PDSCH).

Specifically, the codewords may be converted into scrambled bit sequences by the scramblers **1010**. Scramble sequences used for scrambling may be generated based on an initialization value, and the initialization value may include ID information of a wireless device. The scrambled bit sequences may be modulated to modulation symbol sequences by the modulators **1020**. A modulation scheme may include $\pi/2$ -Binary Phase Shift Keying ($\pi/2$ -BPSK), m-Phase Shift Keying (m-PSK), and m-Quadrature Amplitude Modulation (m-QAM). Complex modulation symbol sequences may be mapped to one or more transport layers by the layer mapper **1030**. Modulation symbols of each transport layer may be mapped (precoded) to corresponding antenna port(s) by the precoder **1040**. Outputs z of the precoder **1040** may be obtained by multiplying outputs y of the layer mapper **1030** by an $N \times M$ precoding matrix W . Herein, N is the number of antenna ports and M is the number of transport layers. The precoder **1040** may perform precoding after performing transform precoding (e.g., DFT) for complex modulation symbols. Alternatively, the precoder **1040** may perform precoding without performing transform precoding.

The resource mappers **1050** may map modulation symbols of each antenna port to time-frequency resources. The time-frequency resources may include a plurality of symbols (e.g., a CP-OFDM symbols and DFT-s-OFDM symbols) in the time domain and a plurality of subcarriers in the frequency domain. The signal generators **1060** may generate radio signals from the mapped modulation symbols and the generated radio signals may be transmitted to other devices through each antenna. For this purpose, the signal generators **1060** may include Inverse Fast Fourier Transform (IFFT) modules, Cyclic Prefix (CP) inserters, Digital-to-Analog Converters (DACs), and frequency up-converters.

Signal processing procedures for a signal received in the wireless device may be configured in a reverse manner of the signal processing procedures **1010** to **1060** of FIG. 25. For example, the wireless devices (e.g., **100** and **200** of FIG. 24) may receive radio signals from the exterior through the antenna ports/transceivers. The received radio signals may be converted into baseband signals through signal restorers. To this end, the signal restorers may include frequency downlink converters, Analog-to-Digital Converters (ADCs), CP remover, and Fast Fourier Transform (FFT) modules. Next, the baseband signals may be restored to codewords through a resource demapping procedure, a postcoding procedure, a demodulation processor, and a descrambling procedure. The codewords may be restored to original information blocks through decoding. Therefore, a signal processing circuit (not illustrated) for a reception signal may include signal restorers, resource demappers, a postcoder, demodulators, descramblers, and decoders.

FIG. 26 illustrates another example of a wireless device applied to the present disclosure. The wireless device may be implemented in various forms according to a use-case/service (refer to FIG. 23 and FIGS. 27 to 32).

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Referring to FIG. 26, wireless devices **100** and **200** may correspond to the wireless devices **100** and **200** of FIG. 24 and may be configured by various elements, components, units/portions, and/or modules. For example, each of the wireless devices **100** and **200** may include a communication unit **110**, a control unit **120**, a memory unit **130**, and additional components **140**. The communication unit may include a communication circuit **112** and transceiver(s) **114**. For example, the communication circuit **112** may include the one or more processors **102** and **202** and/or the one or more memories **104** and **204** of FIG. 24. For example, the transceiver(s) **114** may include the one or more transceivers **106** and **206** and/or the one or more antennas **108** and **208** of FIG. 24. The control unit **120** is electrically connected to the communication unit **110**, the memory **130**, and the additional components **140** and controls overall operation of the wireless devices. For example, the control unit **120** may control an electric/mechanical operation of the wireless device based on programs/code/commands/information stored in the memory unit **130**. The control unit **120** may transmit the information stored in the memory unit **130** to the exterior (e.g., other communication devices) via the communication unit **110** through a wireless/wired interface or store, in the memory unit **130**, information received through the wireless/wired interface from the exterior (e.g., other communication devices) via the communication unit **110**.

The additional components **140** may be variously configured according to types of wireless devices. For example, the additional components **140** may include at least one of a power unit/battery, input/output (I/O) unit, a driving unit, and a computing unit. The wireless device may be implemented in the form of, without being limited to, the robot (**100a** of FIG. 23), the vehicles (**100b-1** and **100b-2** of FIG. 23), the XR device (**100c** of FIG. 23), the hand-held device (**100d** of FIG. 23), the home appliance (**100e** of FIG. 23), the IoT device (**100f** of FIG. 23), a digital broadcast terminal, a hologram device, a public safety device, an MTC device, a medicine device, a fintech device (or a finance device), a security device, a climate/environment device, the AI server/device (**400** of FIG. 23), the BSs (**200** of FIG. 23), a network node, etc. The wireless device may be used in a mobile or fixed place according to a use-example/service.

In FIG. 26, the entirety of the various elements, components, units/portions, and/or modules in the wireless devices **100** and **200** may be connected to each other through a wired interface or at least a part thereof may be wirelessly connected through the communication unit **110**. For example, in each of the wireless devices **100** and **200**, the control unit **120** and the communication unit **110** may be connected by wire and the control unit **120** and first units (e.g., **130** and **140**) may be wirelessly connected through the communication unit **110**. Each element, component, unit/portion, and/or module within the wireless devices **100** and **200** may further include one or more elements. For example, the control unit **120** may be configured by a set of one or more processors. As an example, the control unit **120** may be configured by a set of a communication control processor, an application processor, an Electronic Control Unit (ECU), a graphical processing unit, and a memory control processor. As another example, the memory **130** may be configured by a Random Access Memory (RAM), a Dynamic RAM (DRAM), a Read Only Memory (ROM), a flash memory, a volatile memory, a non-volatile memory, and/or a combination thereof.

Hereinafter, an example of implementing FIG. 26 will be described in detail with reference to the drawings.

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FIG. 27 illustrates a hand-held device applied to the present disclosure. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smart-watch or a smartglasses), or a portable computer (e.g., a notebook). The hand-held device may be referred to as a mobile station (MS), a user terminal (UT), a Mobile Subscriber Station (MSS), a Subscriber Station (SS), an Advanced Mobile Station (AMS), or a Wireless Terminal (WT).

Referring to FIG. 27, a hand-held device **100** may include an antenna unit **108**, a communication unit **110**, a control unit **120**, a memory unit **130**, a power supply unit **140a**, an interface unit **140b**, and an I/O unit **140c**. The antenna unit **108** may be configured as a part of the communication unit **110**. Blocks **110** to **130/140a** to **140c** correspond to the blocks **110** to **130/140** of FIG. 26, respectively.

The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from other wireless devices or BSs. The control unit **120** may perform various operations by controlling constituent elements of the hand-held device **100**. The control unit **120** may include an Application Processor (AP). The memory unit **130** may store data/parameters/programs/code/commands needed to drive the hand-held device **100**. The memory unit **130** may store input/output data/information. The power supply unit **140a** may supply power to the hand-held device **100** and include a wired/wireless charging circuit, a battery, etc. The interface unit **140b** may support connection of the hand-held device **100** to other external devices. The interface unit **140b** may include various ports (e.g., an audio I/O port and a video I/O port) for connection with external devices. The I/O unit **140c** may input or output video information/signals, audio information/signals, data, and/or information input by a user. The I/O unit **140c** may include a camera, a microphone, a user input unit, a display unit **140d**, a speaker, and/or a haptic module.

As an example, in the case of data communication, the I/O unit **140c** may acquire information/signals (e.g., touch, text, voice, images, or video) input by a user and the acquired information/signals may be stored in the memory unit **130**. The communication unit **110** may convert the information/signals stored in the memory into radio signals and transmit the converted radio signals to other wireless devices directly or to a BS. The communication unit **110** may receive radio signals from other wireless devices or the BS and then restore the received radio signals into original information/signals. The restored information/signals may be stored in the memory unit **130** and may be output as various types (e.g., text, voice, images, video, or haptic) through the I/O unit **140c**.

FIG. 28 illustrates a vehicle or an autonomous driving vehicle applied to the present disclosure. The vehicle or autonomous driving vehicle may be implemented by a mobile robot, a car, a train, a manned/unmanned Aerial Vehicle (AV), a ship, etc.

Referring to FIG. 28, a vehicle or autonomous driving vehicle **100** may include an antenna unit **108**, a communication unit **110**, a control unit **120**, a driving unit **140a**, a power supply unit **140b**, a sensor unit **140c**, and an autonomous driving unit **140d**. The antenna unit **108** may be configured as a part of the communication unit **110**. The blocks **110/130/140a** to **140d** correspond to the blocks **110/130/140** of FIG. 26, respectively.

The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from external devices such as other vehicles, BSs (e.g., gNBs and road side units), and servers. The control unit **120** may perform

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various operations by controlling elements of the vehicle or the autonomous driving vehicle **100**. The control unit **120** may include an Electronic Control Unit (ECU). The driving unit **140a** may cause the vehicle or the autonomous driving vehicle **100** to drive on a road. The driving unit **140a** may include an engine, a motor, a powertrain, a wheel, a brake, a steering device, etc. The power supply unit **140b** may supply power to the vehicle or the autonomous driving vehicle **100** and include a wired/wireless charging circuit, a battery, etc. The sensor unit **140c** may acquire a vehicle state, ambient environment information, user information, etc. The sensor unit **140c** may include an Inertial Measurement Unit (IMU) sensor, a collision sensor, a wheel sensor, a speed sensor, a slope sensor, a weight sensor, a heading sensor, a position module, a vehicle forward/backward sensor, a battery sensor, a fuel sensor, a tire sensor, a steering sensor, a temperature sensor, a humidity sensor, an ultrasonic sensor, an illumination sensor, a pedal position sensor, etc. The autonomous driving unit **140d** may implement technology for maintaining a lane on which a vehicle is driving, technology for automatically adjusting speed, such as adaptive cruise control, technology for autonomously driving along a determined path, technology for driving by automatically setting a path if a destination is set, and the like.

For example, the communication unit **110** may receive map data, traffic information data, etc. from an external server. The autonomous driving unit **140d** may generate an autonomous driving path and a driving plan from the obtained data. The control unit **120** may control the driving unit **140a** such that the vehicle or the autonomous driving vehicle **100** may move along the autonomous driving path according to the driving plan (e.g., speed/direction control). In the middle of autonomous driving, the communication unit **110** may aperiodically/periodically acquire recent traffic information data from the external server and acquire surrounding traffic information data from neighboring vehicles. In the middle of autonomous driving, the sensor unit **140c** may obtain a vehicle state and/or surrounding environment information. The autonomous driving unit **140d** may update the autonomous driving path and the driving plan based on the newly obtained data/information. The communication unit **110** may transfer information about a vehicle position, the autonomous driving path, and/or the driving plan to the external server. The external server may predict traffic information data using AI technology, etc., based on the information collected from vehicles or autonomous driving vehicles and provide the predicted traffic information data to the vehicles or the autonomous driving vehicles.

FIG. 29 illustrates a vehicle applied to the present disclosure. The vehicle may be implemented as a transport means, an aerial vehicle, a ship, etc.

Referring to FIG. 29, a vehicle **100** may include a communication unit **110**, a control unit **120**, a memory unit **130**, an I/O unit **140a**, and a positioning unit **140b**. Herein, the blocks **110** to **130/140a** and **140b** correspond to blocks **110** to **130/140** of FIG. 26.

The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from external devices such as other vehicles or BSs. The control unit **120** may perform various operations by controlling constituent elements of the vehicle **100**. The memory unit **130** may store data/parameters/programs/code/commands for supporting various functions of the vehicle **100**. The I/O unit **140a** may output an AR/VR object based on information within the memory unit **130**. The I/O unit **140a** may include an HUD. The positioning unit **140b** may acquire information about

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the position of the vehicle **100**. The position information may include information about an absolute position of the vehicle **100**, information about the position of the vehicle **100** within a traveling lane, acceleration information, and information about the position of the vehicle **100** from a neighboring vehicle. The positioning unit **140b** may include a GPS and various sensors.

As an example, the communication unit **110** of the vehicle **100** may receive map information and traffic information from an external server and store the received information in the memory unit **130**. The positioning unit **140b** may obtain the vehicle position information through the GPS and various sensors and store the obtained information in the memory unit **130**. The control unit **120** may generate a virtual object based on the map information, traffic information, and vehicle position information and the I/O unit **140a** may display the generated virtual object in a window in the vehicle (**1410** and **1420**). The control unit **120** may determine whether the vehicle **100** normally drives within a traveling lane, based on the vehicle position information. If the vehicle **100** abnormally exits from the traveling lane, the control unit **120** may display a warning on the window in the vehicle through the I/O unit **140a**. In addition, the control unit **120** may broadcast a warning message regarding driving abnormality to neighboring vehicles through the communication unit **110**. According to situation, the control unit **120** may transmit the vehicle position information and the information about driving/vehicle abnormality to related organizations.

FIG. 30 illustrates an XR device applied to the present disclosure. The XR device may be implemented by an HMD, an HUD mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance, a digital signage, a vehicle, a robot, etc.

Referring to FIG. 30, an XR device **100a** may include a communication unit **110**, a control unit **120**, a memory unit **130**, an I/O unit **140a**, a sensor unit **140b**, and a power supply unit **140c**. Herein, the blocks **110** to **130/140a** to **140c** correspond to the blocks **110** to **130/140** of FIG. 26, respectively.

The communication unit **110** may transmit and receive signals (e.g., media data and control signals) to and from external devices such as other wireless devices, hand-held devices, or media servers. The media data may include video, images, and sound. The control unit **120** may perform various operations by controlling constituent elements of the XR device **100a**. For example, the control unit **120** may be configured to control and/or perform procedures such as video/image acquisition, (video/image) encoding, and metadata generation and processing. The memory unit **130** may store data/parameters/programs/code/commands needed to drive the XR device **100a**/generate XR object. The I/O unit **140a** may obtain control information and data from the exterior and output the generated XR object. The I/O unit **140a** may include a camera, a microphone, a user input unit, a display unit, a speaker, and/or a haptic module. The sensor unit **140b** may obtain an XR device state, surrounding environment information, user information, etc. The sensor unit **140b** may include a proximity sensor, an illumination sensor, an acceleration sensor, a magnetic sensor, a gyro sensor, an inertial sensor, an RGB sensor, an IR sensor, a fingerprint recognition sensor, an ultrasonic sensor, a light sensor, a microphone and/or a radar. The power supply unit **140c** may supply power to the XR device **100a** and include a wired/wireless charging circuit, a battery, etc.

For example, the memory unit **130** of the XR device **100a** may include information (e.g., data) needed to generate the

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XR object (e.g., an AR/VR/MR object). The I/O unit **140a** may receive a command for manipulating the XR device **100a** from a user and the control unit **120** may drive the XR device **100a** according to a driving command of a user. For example, when a user desires to watch a film or news through the XR device **100a**, the control unit **120** transmits content request information to another device (e.g., a hand-held device **100b**) or a media server through the communication unit **130**. The communication unit **130** may download/stream content such as films or news from another device (e.g., the hand-held device **100b**) or the media server to the memory unit **130**. The control unit **120** may control and/or perform procedures such as video/image acquisition, (video/image) encoding, and metadata generation/processing with respect to the content and generate/output the XR object based on information about a surrounding space or a real object obtained through the I/O unit **140a**/sensor unit **140b**.

The XR device **100a** may be wirelessly connected to the hand-held device **100b** through the communication unit **110** and the operation of the XR device **100a** may be controlled by the hand-held device **100b**. For example, the hand-held device **100b** may operate as a controller of the XR device **100a**. To this end, the XR device **100a** may obtain information about a 3D position of the hand-held device **100b** and generate and output an XR object corresponding to the hand-held device **100b**.

FIG. 31 illustrates a robot applied to the present disclosure. The robot may be categorized into an industrial robot, a medical robot, a household robot, a military robot, etc., according to a used purpose or field.

Referring to FIG. 31, a robot **100** may include a communication unit **110**, a control unit **120**, a memory unit **130**, an I/O unit **140a**, a sensor unit **140b**, and a driving unit **140c**. Herein, the blocks **110** to **130/140a** to **140c** correspond to the blocks **110** to **130/140** of FIG. 26, respectively.

The communication unit **110** may transmit and receive signals (e.g., driving information and control signals) to and from external devices such as other wireless devices, other robots, or control servers. The control unit **120** may perform various operations by controlling constituent elements of the robot **100**. The memory unit **130** may store data/parameters/programs/code/commands for supporting various functions of the robot **100**. The I/O unit **140a** may obtain information from the exterior of the robot **100** and output information to the exterior of the robot **100**. The I/O unit **140a** may include a camera, a microphone, a user input unit, a display unit, a speaker, and/or a haptic module. The sensor unit **140b** may obtain internal information of the robot **100**, surrounding environment information, user information, etc. The sensor unit **140b** may include a proximity sensor, an illumination sensor, an acceleration sensor, a magnetic sensor, a gyro sensor, an inertial sensor, an IR sensor, a fingerprint recognition sensor, an ultrasonic sensor, a light sensor, a microphone, a radar, etc. The driving unit **140c** may perform various physical operations such as movement of robot joints. In addition, the driving unit **140c** may cause the robot **100** to travel on the road or to fly. The driving unit **140c** may include an actuator, a motor, a wheel, a brake, a propeller, etc.

FIG. 32 illustrates an AI device applied to the present disclosure. The AI device may be implemented by a fixed device or a mobile device, such as a TV, a projector, a smartphone, a PC, a notebook, a digital broadcast terminal, a tablet PC, a wearable device, a Set Top Box (STB), a radio, a washing machine, a refrigerator, a digital signage, a robot, a vehicle, etc.

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Referring to FIG. 32, an AI device **100** may include a communication unit **110**, a control unit **120**, a memory unit **130**, an I/O unit **140a/140b**, a learning processor unit **140c**, and a sensor unit **140d**. The blocks **110** to **130/140a** to **140d** correspond to blocks **110** to **130/140** of FIG. 26, respectively.

The communication unit **110** may transmit and receive wired/radio signals (e.g., sensor information, user input, learning models, or control signals) to and from external devices such as other AI devices (e.g., **100x**, **200**, or **400** of FIG. 23) or an AI server (e.g., **400** of FIG. 23) using wired/wireless communication technology. To this end, the communication unit **110** may transmit information within the memory unit **130** to an external device and transmit a signal received from the external device to the memory unit **130**.

The control unit **120** may determine at least one feasible operation of the AI device **100**, based on information which is determined or generated using a data analysis algorithm or a machine learning algorithm. The control unit **120** may perform an operation determined by controlling constituent elements of the AI device **100**. For example, the control unit **120** may request, search, receive, or use data of the learning processor unit **140c** or the memory unit **130** and control the constituent elements of the AI device **100** to perform a predicted operation or an operation determined to be preferred among at least one feasible operation. The control unit **120** may collect history information including the operation contents of the AI device **100** and operation feedback by a user and store the collected information in the memory unit **130** or the learning processor unit **140c** or transmit the collected information to an external device such as an AI server (**400** of FIG. 23). The collected history information may be used to update a learning model.

The memory unit **130** may store data for supporting various functions of the AI device **100**. For example, the memory unit **130** may store data obtained from the input unit **140a**, data obtained from the communication unit **110**, output data of the learning processor unit **140c**, and data obtained from the sensor unit **140d**. The memory unit **130** may store control information and/or software code needed to operate/drive the control unit **120**.

The input unit **140a** may acquire various types of data from the exterior of the AI device **100**. For example, the input unit **140a** may acquire learning data for model learning, and input data to which the learning model is to be applied. The input unit **140a** may include a camera, a microphone, and/or a user input unit. The output unit **140b** may generate output related to a visual, auditory, or tactile sense. The output unit **140b** may include a display unit, a speaker, and/or a haptic module. The sensing unit **140d** may obtain at least one of internal information of the AI device **100**, surrounding environment information of the AI device **100**, and user information, using various sensors. The sensor unit **140d** may include a proximity sensor, an illumination sensor, an acceleration sensor, a magnetic sensor, a gyro sensor, an inertial sensor, an RGB sensor, an IR sensor, a fingerprint recognition sensor, an ultrasonic sensor, a light sensor, a microphone, and/or a radar.

The learning processor unit **140c** may learn a model consisting of artificial neural networks, using learning data. The learning processor unit **140c** may perform AI processing together with the learning processor unit of the AI server (**400** of FIG. 23). The learning processor unit **140c** may process information received from an external device through the communication unit **110** and/or information stored in the memory unit **130**. In addition, an output value

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of the learning processor unit **140c** may be transmitted to the external device through the communication unit **110** and may be stored in the memory unit **130**.

Here, the wireless communication technology implemented in the wireless device (**100, 200**) of the present disclosure may include LTE, NR, and 6G as well as Narrowband Internet of Things for low power communication. At this time, for example, NB-IoT technology may be an example of LPWAN (Low Power Wide Area Network) technology, and may be implemented in standards such as LTE Cat NB1 and/or LTE Cat NB2. not. Additionally or alternatively, the wireless communication technology implemented in the wireless device (**100, 200**) of the present disclosure may perform communication based on LTE-M technology. At this time, as an example, LTE-M technology may be an example of LPWAN technology, and may be called various names such as eMTC (enhanced machine type communication). For example, LTE-M technologies are 1) LTE CAT 0, 2) LTE Cat M1, 3) LTE Cat M2, 4) LTE non-BL (non-Bandwidth Limited), 5) LTE-MTC, 6) LTE Machine Type Communication, and/or 7) It may be implemented in at least one of various standards such as LTE M, and is not limited to the above-mentioned names. Additionally or alternatively, the wireless communication technology implemented in the wireless device (**100, 200**) of the present disclosure is at least one of ZigBee, Bluetooth, and Low Power Wide Area Network (LPWAN) considering low power communication. It may include any one, and is not limited to the above-mentioned names. For example, ZigBee technology can generate personal area networks (PANs) related to small/low-power digital communication based on various standards such as IEEE 802.15.4, and can be called various names.

The above disclosure can be implemented as computer readable code on a medium on which a program is recorded. The computer-readable medium includes all types of recording devices in which data that can be read by a computer system is stored. Examples of computer-readable media include Hard Disk Drive (HDD), Solid State Disk (SSD), Silicon Disk Drive (SDD), ROM, RAM, CD-ROM, magnetic tape, floppy disk, optical data storage device, etc., and also includes those implemented in the form of a carrier wave (eg, transmission over the Internet). Accordingly, the above detailed description should not be construed as limiting in all respects and should be considered illustrative. The scope of this disclosure should be determined by reasonable interpretation of the appended claims, and all changes within the equivalent scope of this disclosure are included in the scope of this disclosure.

The invention claimed is:

1. A method, by a terminal, of performing federated learning in a wireless communication system, the method comprising:

receiving, from a server, a reference signal for channel state information (CSI);
obtaining the CSI based on the reference signal;
obtaining a second flatten vector having gradient values in which signs are aligned, by repeating gradient values included in a first flatten vector and applying clipping and bias values; and

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transmitting the second flatten vector together with the CSI to the server in the form of AirComp.

2. The method of claim 1, wherein the bias value is determined by the number of terminals participating in federated learning and statistical characteristics of the gradients of the first flatten vector.

3. The method of claim 2, wherein the bias value is determined using a value defined in advance as a table in a transceiver.

4. The method of claim 1, wherein the CSI is subjected to phase compensation.

5. The method of claim 1,

wherein the signs include a positive sign and a negative sign,

wherein values having a negative sign in a first partial vector among the gradient values included in the second flatten vector are clipped, and a negative bias is applied to the partial vector, and

wherein values having a positive sign in a second partial vector among the gradient values included in the second flatten vector are clipped, and a positive bias is applied to the partial vector.

6. The method of claim 1, wherein a signal transmitted in the transmitting in the form of AirComp has a predetermined transmission power.

7. The method of claim 1, further comprising:

grouping at least one terminal participating in learning among a plurality of random terminals based on the CSI.

8. The method of claim 7, wherein a signal transmitted by at least one terminal belonging to the same group has contiguous reception power sensitivity.

9. A method, by a server, of performing federated learning in a wireless communication system, the method comprising:

transmitting, to a plurality of terminals, a reference signal for channel state information (CSI);

receiving, from the plurality of terminals, the CSI based on the reference signal;

receiving gradient signals for federated learning transmitted in an AirComp form from the plurality of terminals, a signal includes the CSI and a second flatten vector;

obtaining an aggregated gradient based on the received gradient signals; and

updating parameters of the global model based on the aggregated gradient.

10. The method of claim 9, wherein the obtaining of the aggregated gradient includes calculating one aggregated gradient by overlapping the gradient signals for each entry.

11. The method of claim 10, wherein a bias value is eliminated and not present in the aggregated gradient as a result of overlapping for each entry.

12. A non-transitory computer-readable recording medium having a program for executing the method of claim 1 in a computer system recorded thereon.

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